USER MANUAL

POTENTIALBOX

A groovebox for Windows, played by the Arturia BeatStep Pro

Version 1.8.0 · drum sampler · two synths · 16 performance macros · ASIO



by PotentialBreaks

Free and open source · MIT licence

1 Welcome

PotentialBox turns your Arturia BeatStep Pro into a complete groovebox. The BeatStep Pro is the sequencer and the controller, and PotentialBox makes the sound on your Windows PC:

- **Sequencer 1** plays **Synth 1**, an analog-style mono synth for basslines, acid and leads.
- **Sequencer 2** plays **Synth 2**, a wavetable mono synth for leads, pads and bells.
- The **Drum** sequencer plays a **16-pad drum sampler**, with 9 built-in kits, your own sample folders and a browser for whole drum machine collections.
- Each synth has **2 LFOs, a mod envelope and 8 mod slots**, five filter types, chords, FM, ring mod and sync.
- An **arpeggiator** on each synth, **side-chain ducking**, EQ and a glue compressor in the mix.
- A **Live tab** with hold-to-play punch-in effects, scenes you can morph between, recording to WAV (with stems) and resampling onto a drum pad.
- A **beat generator** with 24 genres, from techno and house to drum & bass, reggae, boom bap, trap and grime. One tap for a new take, and **SEND TO BEATSTEP** records it straight into the BeatStep's drum sequencer.
- A **5-track looper** in the style of a BOSS RC-505: record your mic, guitar or keys and PotentialBox itself, overdub, undo, all played from the BeatStep's pads and locked to its tempo.
- **Step locks**: give any knob a different value on any of the 16 steps. Drums also get chance and "every Nth bar" rules.
- The BeatStep's **16 knobs** control the 16 knobs on whichever PotentialBox page is open.
- The BeatStep's **pads** (on MIDI channel 16) run actions such as random sounds and switching pages.

Everything is pre-wired for the BeatStep Pro's factory settings, so you can plug it in and press play. Every control is a big button you can tap, and everything saves by itself, so there is no typing anywhere in the app.

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2 What you need

YOU NEED	NOTES
A Windows PC	Windows 10 or 11, 64-bit.
An Arturia BeatStep Pro	Connected with USB. Factory MIDI settings work straight away.
Internet, once	The installer downloads Python 3.12 and the audio libraries (about 150 MB). After that, PotentialBox works offline.
Speakers or headphones	Any Windows audio output works.

Nice to have

OPTIONAL	WHY
An audio interface with an ASIO driver	The lowest lag between pressing play and hearing sound. Without one, PotentialBox uses Windows audio (WASAPI), which is still fine for playing.
A folder of drum samples	For example a drum machine collection: one folder per machine, named after the hardware. PotentialBox reads them in place.
Arturia MIDI Control Center	Arturia's free app, to move the BeatStep's Control Mode pads to MIDI channel 16 for pad actions (chapter 5).

3 Installing

1. Download `PotentialBox-1.8.0-windows.zip` from the **Releases** page of the PotentialBox GitHub repository.
2. Right-click the zip and choose **Extract All**. Open the extracted **PotentialBox** folder.
3. Double-click **Install PotentialBox.bat**.
4. Wait for the window to say **DONE**. PotentialBox opens by itself, and there is now a PotentialBox icon on your Desktop and in the Start menu.

> **"Windows protected your PC"?** PotentialBox is free and not code-signed, so Windows may show this the first time. Click **More info**, then **Run anyway**.

What the installer does

- Installs a **private copy of Python 3.12** in your user folder if you don't have it. It does not change any other Python you have.
- Copies PotentialBox to `%LOCALAPPDATA%\PotentialBox` and installs its libraries there: numpy, scipy, sounddevice (with ASIO support), mido and python-rtmidi.
- Lists the ASIO outputs and MIDI inputs it can see, so you know the BeatStep was found.
- Adds the Desktop and Start menu shortcuts, then starts the app.

Updating

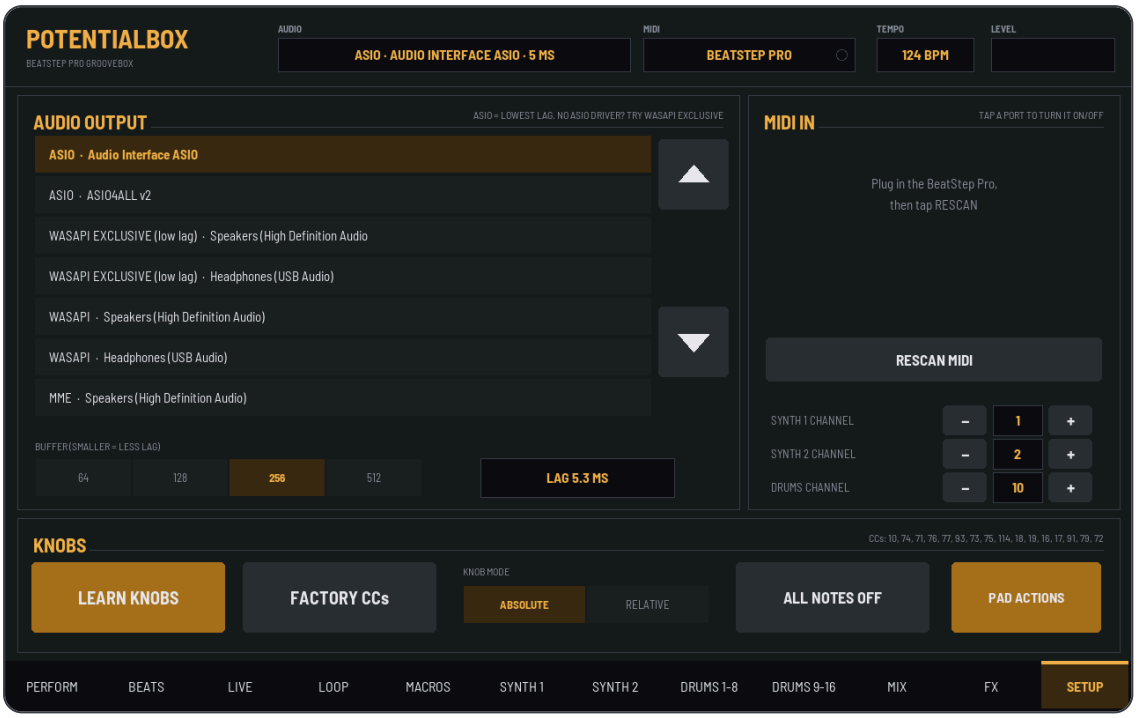
Download the new release and run its **Install PotentialBox.bat**. The installer closes PotentialBox if it is open, updates it, and keeps all your settings, saved sounds and kits. The window title shows the version you are running.

Uninstalling

Open `%LOCALAPPDATA%\PotentialBox`, right-click **uninstall.ps1** and choose **Run with PowerShell**. This removes the app, its shortcuts and your settings. Python 3.12 stays; remove it from Windows Settings > Apps if you like.

4 First start: Setup

Open the SETUP tab (bottom right) the first time you start PotentialBox.



The Setup tab: audio outputs on the left, MIDI on the right, knob settings at the bottom.

Audio output

Tap an output to use it. The list is sorted best first:

OUTPUT TYPE	WHEN TO USE IT
ASIO	Lowest lag. Choose your audio interface's own ASIO driver. PotentialBox picks it automatically the first time, before generic drivers like ASIO4ALL.
WASAPI EXCLUSIVE	Very low lag with no extra driver. Other apps go quiet while PotentialBox uses the output.
WASAPI	Works everywhere, a little more lag. Other apps keep playing.

BUFFER sets the lag. Smaller is faster; if you hear crackles, tap the next size up. The **LAG** readout shows the result in milliseconds. 128 or 256 is a good start.

> **ASIO busy?** Many audio interfaces let only one app use their ASIO driver at a time. If a DAW, FL Studio or Voicemeeter is open, PotentialBox can't get the driver, so it plays through Windows audio and tells you why. Close the other app, then tap the ASIO output again.

If an output ever freezes, PotentialBox remembers it, skips it next time and marks it **(DID NOT RESPOND)** in the list. Tapping it again tries it once more.

MIDI in

PotentialBox finds the BeatStep Pro by itself and listens to every port it offers. Tap a port to switch it off or on, and **RESCAN MIDI** after plugging in. The **channel** boxes set which MIDI channel plays each instrument (factory: Synth 1 = 1, Synth 2 = 2, Drums = 10).

5 Setting up the BeatStep Pro

The sequencers

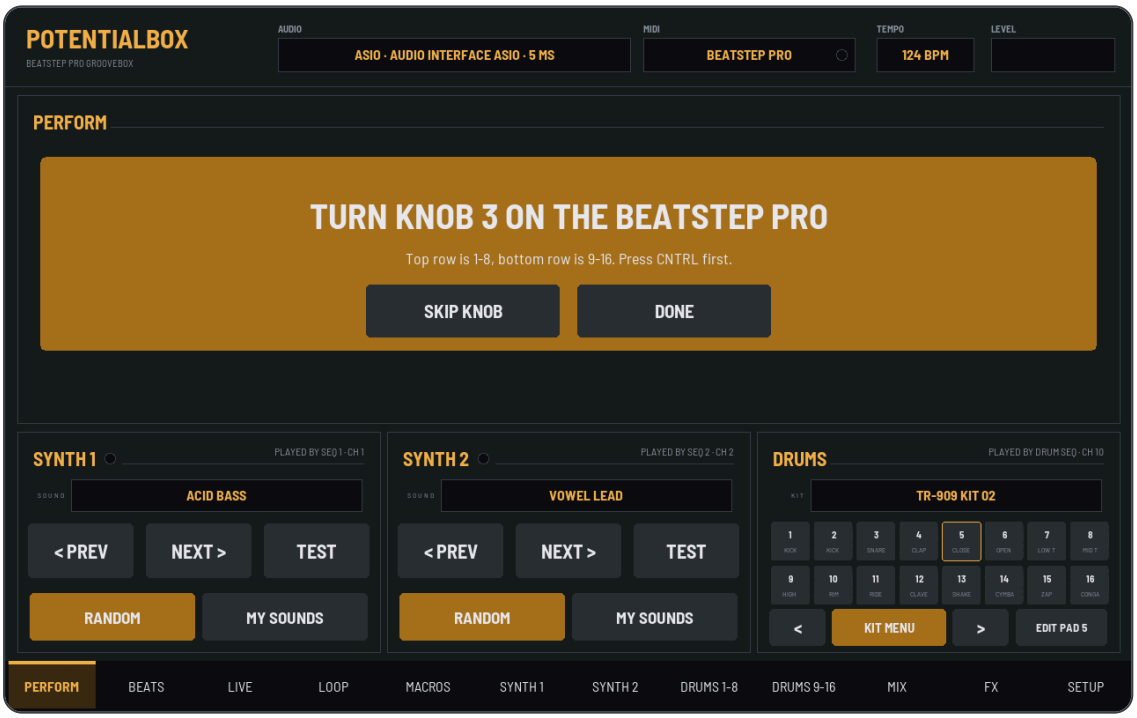
Nothing to set up: on factory settings Sequencer 1 sends on channel 1, Sequencer 2 on channel 2 and the Drum sequencer on channel 10 with notes 36-51, which is exactly what PotentialBox listens to.

- Press **PLAY** on the BeatStep Pro. PotentialBox follows its tempo (shown in the header), so the delay always lands on the beat.
- **Stop** also sends All Notes Off, so no note hangs.
- **Tied and slid steps** glide from note to note. Set the slide time with the synth's Glide knob.
- **Swing, randomness and probability** are the BeatStep's own: they change the pattern before PotentialBox plays it.

The knobs: Control Mode

The knobs only send to PotentialBox in **Control Mode**. Press `CONTROL MODE` on the BeatStep Pro; the sequencers keep playing. Turn a knob and the matching knob on screen lights up.

If a knob does nothing, open `SETUP`, tap `LEARN KNOBS` and turn knobs 1 to 16 in order (top row first). **FACTORY CCs** undoes this.



LEARN KNOBS: turn each BeatStep knob when asked.

If your knobs are set to send **relative** values in MIDI Control Center, switch **KNOB MODE** to **RELATIVE**.

The pads: channel 16

To use the pads for pad actions (chapter 19), set the BeatStep Pro's Control Mode pads to **MIDI channel 16** in Arturia's free **MIDI Control Center** app. Out of the box, pads 1-16 send notes 36-51, which PotentialBox already expects. Any other

notes or CCs work too: assign them in SETUP > PAD ACTIONS.

6 The screen



The Perform tab with the BeatStep Pro connected.

Header

- **AUDIO**: the output in use and its lag. Pop-up messages also appear here for a moment.
- **MIDI**: whether the BeatStep Pro is connected. The light flashes when MIDI arrives.
- **TEMPO**: the BeatStep's tempo. **LEVEL**: the output meter (it turns red if you push it into the limiter).

Tabs

The tabs along the bottom choose what the 16 knobs control: Perform, Beats, Live, Loop, Macros, Synth 1, Synth 2, Drums 1-8, Drums 9-16, Mix, FX, Setup. The knob area is laid out exactly like the BeatStep's knobs: top row 1-8, bottom row 9-16, in labelled groups.

Some tabs have more than one page. The page buttons sit at the top of the knob area; tapping the tab again also goes to the next page:

TAB	PAGES
Synth 1	MAIN · OSC+ · MOD · ARP+
Synth 2	MAIN · OSC+ · MOD · ARP+
Drums 1-8	MAIN · MIX · GROOVE
Drums 9-16	MAIN · MIX · GROOVE
Mix	MAIN · EQ

Using a knob on screen

- **Tap the left or right half** of a knob to step it down or up (no dragging needed).
- Or **drag** up and down, or use the **mouse wheel**.
- A knob that a macro is moving shows its value in amber, with a small ring where your own setting is.
- **STEP LOCKS** (top right of the knob area) lets a knob have a different value on each step: chapter 17.
- A small amber **L** on a knob means it has step locks.

The three panels

Under the knobs, **Synth 1**, **Synth 2** and **Drums** each have their own panel. The light next to a synth's name shows when it is playing; drum pads light up when they hit. The buttons are explained in chapters 10 and 11.

7 Synth 1

An analog-style mono synth: two oscillators that blend from saw to square, a sub oscillator, a resonant low-pass filter with its own envelope, and drive. It is played by **Sequencer 1**.

The Synth 1 tab has three pages: **MAIN**, **OSC+** and **MOD**. Tap them at the top of the knob area, or tap the Synth 1 tab again (or press its BeatStep pad) to go to the next page.

> **Start again in one tap:** each page has an amber INIT button beside the page buttons. **INIT SYNTH 1** on MAIN resets the whole synth (all three pages). **INIT OSC+ PAGE** resets only the OSC+ controls (the volume is kept). **INIT MODULATION** resets the LFOs, mod envelope and all 8 mod slots. After any of them, **UNDO INIT** appears for a few seconds to bring your settings back. Synth 2 has the same buttons.



Synth 1, MAIN page.

MAIN page

KNOB	CONTROL	WHAT IT DOES
1	**Cutoff**	How bright the sound is. Low = dark and muffled, high = open and bright.
2	**Resonance**	Boosts the frequencies around the cutoff. High settings squelch and whistle.
3	**Filter env**	How far each note opens the filter. The heart of acid and pluck sounds.
4	**Filter decay**	How quickly the filter closes again after each note.
5	**Saw > Square**	Blends the oscillator from a saw wave (buzzy) to a square wave (hollow).
6	**Pulse width**	Shape of the square wave. 50% is hollow, small values are thin and nasal.
7	**Detune**	Detunes the second oscillator for a thicker sound.
8	**Sub osc**	Adds a sine wave one octave down for weight.
9	**Attack**	How long the note takes to fade in.
10	**Decay**	How long the note takes to fall to the sustain level.

KNOB	CONTROL	WHAT IT DOES
11	**Sustain**	The level the note holds at while the step is held.
12	**Release**	How long the note rings after the step ends.
13	**Glide (slides)**	Slide time between notes on tied or slid BeatStep steps.
14	**Octave**	Moves Synth 1 up or down in octaves.
15	**Drive**	Saturation after the filter: warmth at low settings, grit at high.
16	**Synth 1 volume**	Synth 1's level in the mix.

OSC+ page



Synth 1, OSC+ page.

KNOB	CONTROL	WHAT IT DOES
1	**Osc 2 pitch**	Tunes oscillator 2 in semitones (+7 = a fifth, +12 = an octave). With Hard sync on, it sets the sync tone.
2	**FM (osc 2 > 1)**	Oscillator 2 bends oscillator 1's waveform: metallic, bell-like and growly tones.
3	**Ring mod**	Multiplies the two oscillators together: clangy, bell-like tones.
4	**Hard sync**	Locks oscillator 2 to oscillator 1. Sweep Osc 2 pitch (or modulate it) for the classic sync scream.
5	**Noise**	Adds white noise before the filter.
6	**Chord**	Plays a chord on every step, even though the BeatStep sequences are one note at a time.
7	**Chord voicing**	Close, Open (the middle note goes up an octave) or Wide (spread further).
8	**Pan**	Places the synth left or right in the mix.
9	**Filter type**	LP 24 (smooth, classic), LP 12 (brighter), HP 12 (thin), Band-pass (nasal) or Notch (hollow).
10	**Key tracking**	Higher notes open the filter more, like an acoustic instrument.

KNOB	CONTROL	WHAT IT DOES
11	**Velocity > volume**	How much step velocity changes the volume. 0% makes every step the same level.
12	**Analog drift**	Slow random detuning of each voice, like an old analog synth.
13	**Pitch env**	A quick pitch drop or rise at the start of each note, in semitones. Great for zaps and kicks.
14	**Pitch env time**	How quickly the pitch envelope settles.
15	**Glide mode**	Slides only: glide on tied or slid steps. Always: every new note glides from the last one.
16	**Synth 1 volume**	Synth 1's level in the mix.

MOD page

LFOs, the mod envelope and the 8 mod slot amounts. Chapter 9 explains how modulation works.

KNOB	CONTROL	WHAT IT DOES
1	**LFO 1 shape**	Sine, Triangle, Saw up, Saw down, Square, Random (steps) or Smooth random.
2	**LFO 1 speed**	Speed in Hz when LFO 1 sync is Free (shows Synced otherwise).
3	**LFO 1 sync**	Free, or locked to the BeatStep's tempo: 4 bars down to 1/32, plus triplets.
4	**LFO 2 shape**	Sine, Triangle, Saw up, Saw down, Square, Random (steps) or Smooth random.
5	**LFO 2 speed**	Speed in Hz when LFO 2 sync is Free (shows Synced otherwise).
6	**LFO 2 sync**	Free, or locked to the BeatStep's tempo: 4 bars down to 1/32, plus triplets.
7	**Mod env attack**	How long the mod envelope takes to rise. It restarts on every new note.
8	**Mod env decay**	How long the mod envelope takes to fall back to zero.
9	**Slot 1 amount**	Amount for slot 1. Out of the box: LFO 1 to Pitch. Tap its name to re-route.
10	**Slot 2 amount**	Amount for slot 2. Out of the box: LFO 1 to Cutoff. Tap its name to re-route.
11	**Slot 3 amount**	Amount for slot 3. Out of the box: LFO 2 to Pulse width. Tap its name to re-route.
12	**Slot 4 amount**	Amount for slot 4. Out of the box: Mod env to Cutoff. Tap its name to re-route.
13	**Slot 5 amount**	Amount for slot 5. Out of the box: Velocity to Cutoff. Tap its name to re-route.
14	**Slot 6 amount**	Amount for slot 6. Out of the box: LFO 2 to Pan. Tap its name to re-route.
15	**Slot 7 amount**	Amount for slot 7. Out of the box: Mod env to Pitch. Tap its name to re-route.
16	**Slot 8 amount**	Amount for slot 8. Out of the box: LFO 2 to Volume. Tap its name to re-route.

ARP+ page



Synth 1, ARP+ page.

KNOB	CONTROL	WHAT IT DOES
1	**Arp**	Off, Up, Down, Up-down, Random or As played (the order the notes arrived).
2	**Arp rate**	How fast it plays, locked to the BeatStep's tempo: 1/4, 1/8, 1/8T, 1/16, 1/16T, 1/32.
3	**Arp octaves**	Plays the notes over 1 to 4 octaves.
4	**Arp gate**	How long each arp note lasts. At 100% the notes slide into each other.
5	**Arp swing**	Pushes every second arp note later for a shuffle.
6	**Arp hold**	On: the arp keeps playing after the BeatStep's note ends, until the next new note.
7	**Arp chance**	Below 100%, some arp notes are skipped at random.
8	**Arp accent**	Makes every 2nd, 3rd or 4th arp note louder and the others softer.
9	**Duck Synth 1**	Side-chain duck: Synth 1 dips when the kick plays (chapter 18).
10	**Synth 1 Low**	Low EQ for Synth 1 (bass), -12 to +12 dB.
11	**Synth 1 Mid**	Mid EQ for Synth 1 (around 1 kHz).
12	**Synth 1 High**	High EQ for Synth 1 (treble).
13	**Pan**	Places Synth 1 left or right.
14	**Delay: Synth 1**	Delay send.
15	**Reverb: Synth 1**	Reverb send.
16	**Synth 1 volume**	Synth 1's level in the mix.

USING THE ARPEGGIATOR

- Turn **Arp** on and play a long note on the BeatStep (a tied step or a long gate). The arp plays from it.
- With **Chord** on (OSC+ page), the arp plays the chord's notes one at a time, the classic way to get arpeggios from a one-note sequencer.
- Short BeatStep steps also work: each new step plays the next note of the arp pattern, so a plain 16-step sequence walks through the chord.

- **INIT ARP** turns the arpeggiator off and resets its settings in one tap.

8 Synth 2

A wavetable mono synth: six tables that morph as you move the wave position, an LFO that sweeps it, two detuned unison voices in stereo, and a resonant filter. It is played by **Sequencer 2**.

Like Synth 1, the Synth 2 tab has **MAIN**, **OSC+** and **MOD** pages.



Synth 2, MAIN page.

MAIN page

KNOB	CONTROL	WHAT IT DOES
1	**Wave position**	Where in the wavetable you are. Each table morphs as you turn this.
2	**Env → Position**	How much each note moves the wave position (up or down).
3	**LFO speed**	Speed of the LFO that moves the wave position.
4	**LFO → Position**	How far the LFO moves the wave position.
5	**Cutoff**	How bright the sound is.
6	**Resonance**	Resonance at the cutoff.
7	**Filter env**	How far each note opens the filter.
8	**Wavetable**	Chooses the wavetable: Analog, Vowel, Harmonic, PWM, FM Bell, Digital, FM Synth, E-Organ, E-Piano, Voice, Granular, Video Game, Chiptune, Bitcrushed, Overtones, String Box, Distorted, Raw, Snippets, Pluck, Clavinet, E-Bass, Deep Bass, Flute, Birds, Symmetric, Sine Harm, Theremin, Bright Saw, Linear.
9	**Attack**	Fade-in time.
10	**Decay**	Fall time to the sustain level. Also sets how quickly the filter and wave envelope close.
11	**Sustain**	Held level.
12	**Release**	Ring-out time after the step ends.

KNOB	CONTROL	WHAT IT DOES
13	**Glide (slides)**	Slide time between tied or slid steps.
14	**Octave**	Moves Synth 2 up or down in octaves.
15	**Unison**	Unison: two detuned voices spread left and right.
16	**Synth 2 volume**	Synth 2's level in the mix.

OSC+ page



Synth 2, OSC+ page.

KNOB	CONTROL	WHAT IT DOES
1	**Chord**	Plays a chord on every step, even though the BeatStep sequences are one note at a time.
2	**Chord voicing**	Close, Open (the middle note goes up an octave) or Wide (spread further).
3	**Sub osc**	Adds a sine wave one octave down for weight.
4	**Noise**	Adds white noise before the filter.
5	**Pan**	Places the synth left or right in the mix.
6	**Analog drift**	Slow random detuning of each voice, like an old analog synth.
7	**Pitch env**	A quick pitch drop or rise at the start of each note, in semitones. Great for zaps and kicks.
8	**Pitch env time**	How quickly the pitch envelope settles.
9	**Filter type**	LP 24 (smooth, classic), LP 12 (brighter), HP 12 (thin), Band-pass (nasal) or Notch (hollow).
10	**Key tracking**	Higher notes open the filter more, like an acoustic instrument.
11	**Velocity > volume**	How much step velocity changes the volume. 0% makes every step the same level.
12	**Glide mode**	Slides only: glide on tied or slid steps. Always: every new note glides from the last one.
13	**Wavetable**	Quick access: chooses the wavetable.

KNOB	CONTROL	WHAT IT DOES
14	**Wave position**	Quick access: where in the wavetable you are.
15	**Cutoff**	Quick access: filter cutoff.
16	**Synth 2 volume**	Synth 2's level in the mix.

MOD page

KNOB	CONTROL	WHAT IT DOES
1	**LFO 1 shape**	Sine, Triangle, Saw up, Saw down, Square, Random (steps) or Smooth random.
2	**LFO 1 speed**	Speed in Hz when LFO 1 sync is Free (shows Synced otherwise).
3	**LFO 1 sync**	Free, or locked to the BeatStep's tempo: 4 bars down to 1/32, plus triplets.
4	**LFO 2 shape**	Sine, Triangle, Saw up, Saw down, Square, Random (steps) or Smooth random.
5	**LFO 2 speed**	Speed in Hz when LFO 2 sync is Free (shows Synced otherwise).
6	**LFO 2 sync**	Free, or locked to the BeatStep's tempo: 4 bars down to 1/32, plus triplets.
7	**Mod env attack**	How long the mod envelope takes to rise. It restarts on every new note.
8	**Mod env decay**	How long the mod envelope takes to fall back to zero.
9	**Slot 1 amount**	Amount for slot 1. Out of the box: LFO 1 to Pitch. Tap its name to re-route.
10	**Slot 2 amount**	Amount for slot 2. Out of the box: LFO 1 to Cutoff. Tap its name to re-route.
11	**Slot 3 amount**	Amount for slot 3. Out of the box: LFO 2 to Wave position. Tap its name to re-route.
12	**Slot 4 amount**	Amount for slot 4. Out of the box: Mod env to Wave position. Tap its name to re-route.
13	**Slot 5 amount**	Amount for slot 5. Out of the box: Velocity to Cutoff. Tap its name to re-route.
14	**Slot 6 amount**	Amount for slot 6. Out of the box: LFO 2 to Pan. Tap its name to re-route.
15	**Slot 7 amount**	Amount for slot 7. Out of the box: Mod env to Pitch. Tap its name to re-route.
16	**Slot 8 amount**	Amount for slot 8. Out of the box: LFO 2 to Volume. Tap its name to re-route.

ARP+ page

The same arpeggiator, duck, EQ and output controls as on Synth 1 (chapter 7).

The 30 wavetables

Built in: Analog sine to triangle to saw to square · **Vowel** talking "ah-eh-ee-oh-oo" · **Harmonic** a sweeping band of harmonics · **PWM** pulse width sweep · **FM Bell** metallic FM tones · **Digital** gritty digital spectra.

From the AKWF collection (each morphs through 16 related single-cycle waveforms): FM Synth, E-Organ, E-Piano, Voice, Granular, Video Game, Chiptune, Bitcrushed, Overtones, String Box, Distorted, Raw, Snippets, Pluck, Clavinet, E-

Bass, Deep Bass, Flute, Birds, Symmetric, Sine Harm, Theremin, Bright Saw, Linear.

9 Modulation and chords

Modulation makes a sound move by itself: a wobbling filter, vibrato, a sweeping wavetable, auto-pan. Each synth has its own modulation on its **MOD** page.



Synth 1, MOD page: LFOs and mod envelope on top, the 8 mod slot amounts below.

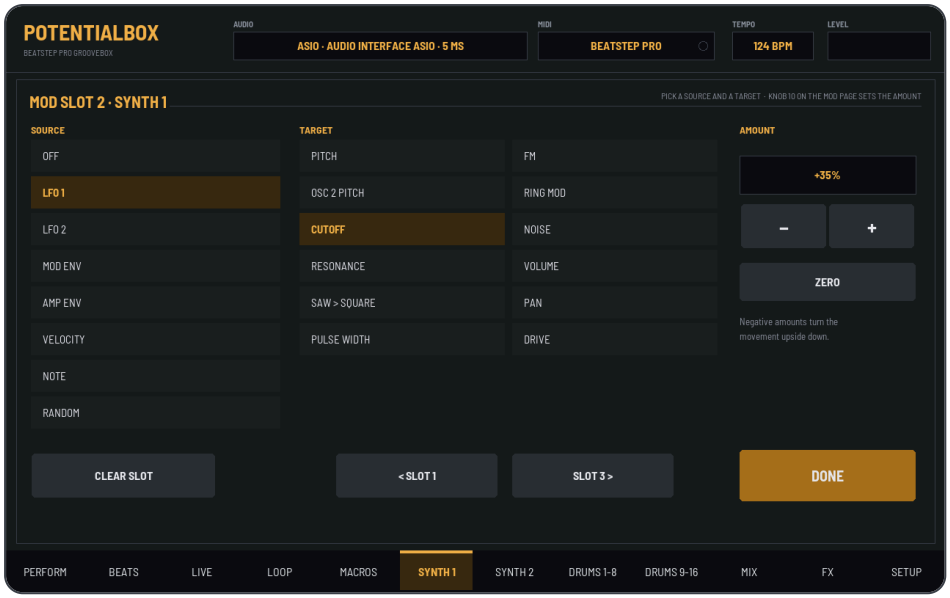
The three movers

- **LFO 1 and LFO 2** are slow oscillators that move things up and down. Shapes: Sine, Triangle, Saw up, Saw down, Square, Random, Smooth random. Set the speed in Hz, or lock them to the BeatStep's tempo with **sync** (4 bars, 2 bars, 1 bar, 1/2, 1/4, 1/8, 1/16, 1/32, 1/4T, 1/8T, 1/16T).
- The **mod envelope** rises and falls once on every new note. Use it for sweeps, zaps and plucks.
- There are also four sources that need no setting up: the **amp envelope**, the step's **velocity**, the **note** itself (higher notes give more) and **Random** (a new value on every note).

Mod slots

Each synth has **8 mod slots**, and each slot connects one source to one target. The bottom row of the MOD page holds the 8 **amounts** (BeatStep knobs 9 to 16), labelled with their routing, for example "LFO 1 > Cutoff".

- All amounts start at **0**, so nothing changes until you turn one up.
- A **negative** amount turns the movement upside down.
- **Tap a slot's name** to choose a different source or target.
- **INIT MODULATION** (top of the MOD page) puts both LFOs, the mod envelope and all 8 slots (routing and amount) back to how they started, in one tap. Changed your mind? The button turns into **UNDO INIT** for a few seconds.



The routing menu: tap a source, tap a target. CLEAR SLOT switches the slot off.

OUT-OF-THE-BOX ROUTINGS

SLOT	SYNTH 1	SYNTH 2
1	LFO 1 > Pitch	LFO 1 > Pitch
2	LFO 1 > Cutoff	LFO 1 > Cutoff
3	LFO 2 > Pulse width	LFO 2 > Wave position
4	Mod env > Cutoff	Mod env > Wave position
5	Velocity > Cutoff	Velocity > Cutoff
6	LFO 2 > Pan	LFO 2 > Pan
7	Mod env > Pitch	Mod env > Pitch
8	LFO 2 > Volume	LFO 2 > Volume

TARGETS

Synth 1: Pitch, Osc 2 pitch, Cutoff, Resonance, Saw > Square, Pulse width, FM, Ring mod, Noise, Volume, Pan, Drive.
Synth 2: Pitch, Cutoff, Resonance, Wave position, Unison, Sub osc, Noise, Volume, Pan.

Recipes

SOUND	HOW
Vibrato	Slot 1 (LFO 1 > Pitch) at about +3%, LFO 1 speed around 5 Hz.
Wobble bass	LFO 1 sync 1/8, slot 2 (LFO 1 > Cutoff) at +40 to +60%, filter type LP 12, some resonance.
Sync scream	Hard sync On, Osc 2 pitch +12, then slot 7 re-routed to Mod env > Osc 2 pitch at +60%.
Auto-pan	Slot 6 (LFO 2 > Pan) at +50%, LFO 2 sync 1/2.

SOUND	HOW
Harder steps brighter	Slot 5 (Velocity > Cutoff) at +30%, then vary step velocities on the BeatStep.
Every note different	Route Random > Cutoff or Random > Wave position at a small amount.
Evolving pad	Synth 2: LFO 2 shape Smooth random, slow speed, slot 3 (LFO 2 > Wave position) at +40%.

Chords

The BeatStep's sequencers play one note at a time, but **Chord** on the OSC+ page turns every step into a chord: Octave, Fifth, Fifth + octave, Major, Minor, Sus2, Sus4, Major 7, Minor 7, Dominant 7, Minor 9, Stack. **Chord voicing** spreads it: Close, Open, Wide. All the chord's notes go through the same filter and envelopes, like a classic paraphonic synth.

> **Heavy patches:** chords on both synths with lots of modulation use more of your PC. If you hear crackles, choose the next BUFFER size up in Setup (256 is a good choice).

NEW PRESETS THAT SHOW THIS OFF

Synth 1: Sync Lead, FM Bass, Wobble Bass, Minor 7 Stab. **Synth 2:** Chord Pad, Chip Lead, Voice Choir, Granular Drone.

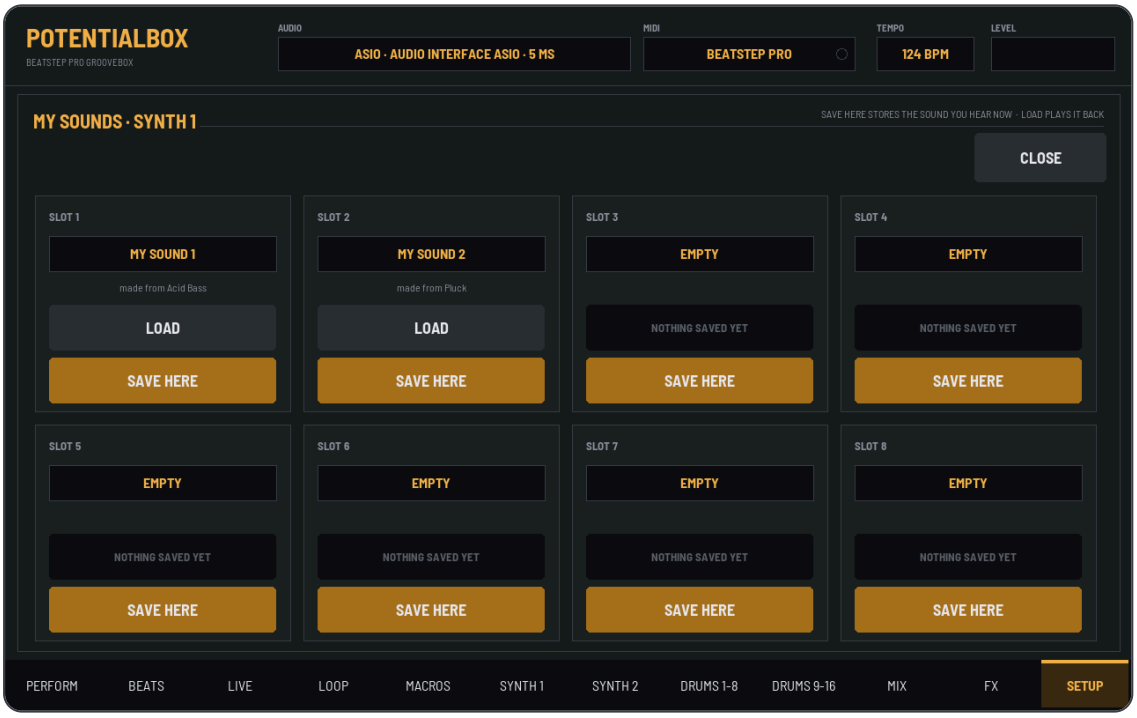
10 Sounds and save slots

The synth panels

- `< PREV` and `NEXT >` step through sounds and play a test note. The list starts with **Init** (a plain starting point), then the presets, then your saved sounds.
- `TEST` plays a note so you can hear the synth without the BeatStep.
- `RANDOM` rolls a brand-new playable sound.
- `MY SOUNDS` opens your save slots.

Synth 1 presets: Acid Bass, Fat Saw Lead, Square Bass, Pluck, Sub Bass, Soft Lead, Sync Lead, FM Bass, Wobble Bass, Minor 7 Stab.
Synth 2 presets: Glass Bell, Vowel Lead, Digital Bass, PWM Strings, Harmonic Sweep, Warm Pad, Chord Pad, Chip Lead, Voice Choir, Granular Drone.

My Sounds



My Sounds: 8 slots for each synth.

- **SAVE HERE** stores the sound you are hearing. No typing: slots are named My Sound 1 to 8 and remember which sound they were made from.
- On a slot that is already full, SAVE HERE asks you to **tap again** within 3 seconds, so one slip can't overwrite a sound.
- **LOAD** plays the saved sound back. Saved sounds also appear in PREV / NEXT.

> **Start from scratch:** pick **Init** with PREV / NEXT for one synth, or tap **RESET ALL TO INIT** on the Mix tab (tap it twice) to reset every knob in the app. Saved sounds, kits and setup are kept.

11 Drums and kits

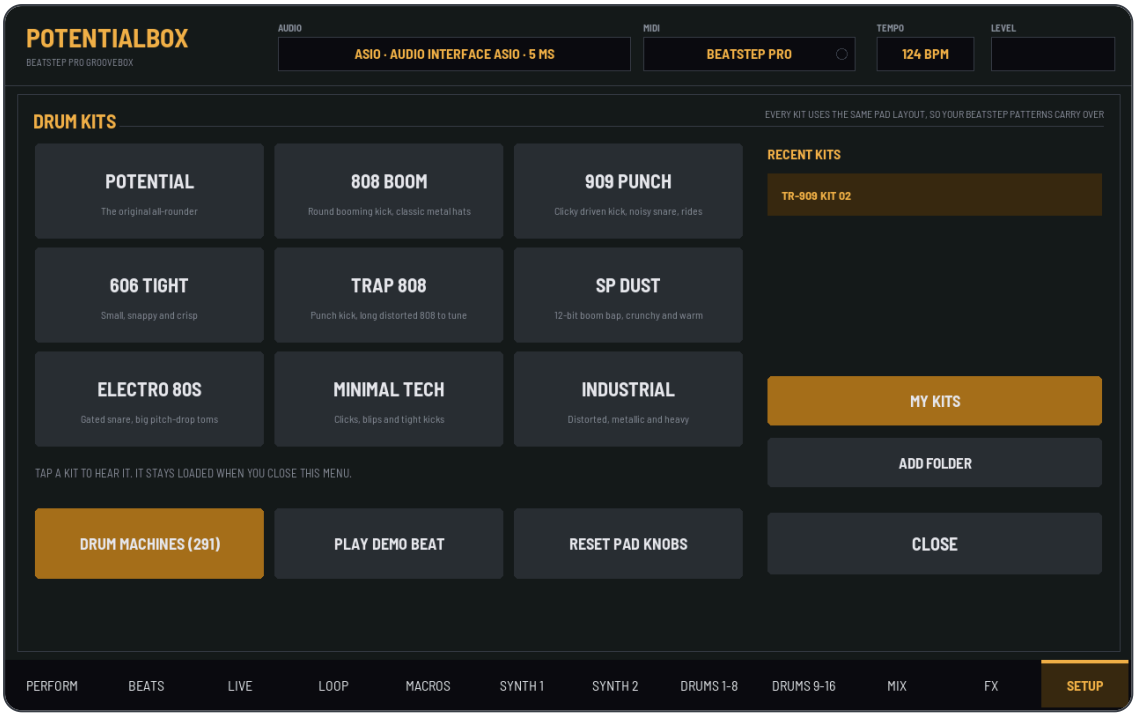
The drum sampler has 16 pads, played by the BeatStep's Drum sequencer (notes 36-51). **Every kit uses the same pad layout**, so a drum pattern keeps making sense when you change kits:

PAD	NOTE	SOUND
1	36	Kick
2	37	Kick 2 (second kick or 808)
3	38	Snare
4	39	Clap
5	40	Closed hat (cuts the open hat)
6	41	Open hat
7	42	Low tom
8	43	Mid tom
9	44	High tom
10	45	Rim
11	46	Percussion (cowbell)
12	47	Percussion (clave)
13	48	Percussion (shaker)
14	49	Cymbal (crash or ride)
15	50	FX
16	51	Percussion (conga)

Pad 5 cuts off pad 6, like a real hi-hat. Tap a pad on screen to hear it and select it (amber outline).

Kit menu

Tap KIT MENU on the drum panel, or < and > to step through kits.



The Kit menu: built-in kits, recent kits, My Kits and the drum machine library.

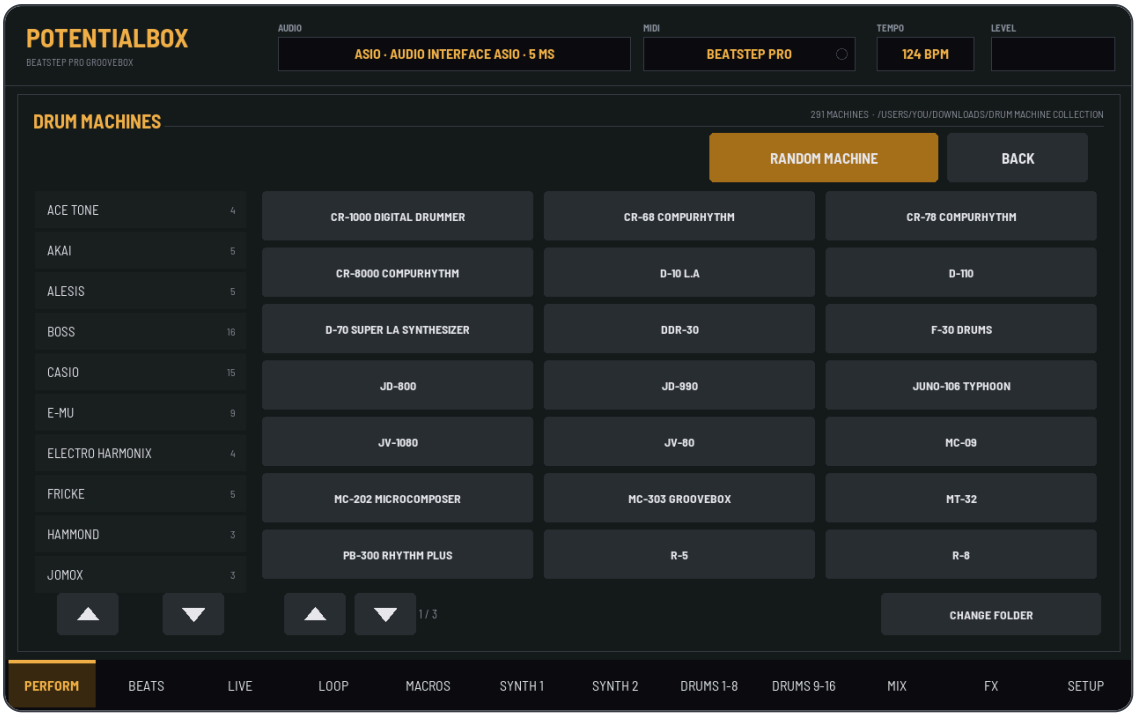
BUILT-IN KITS

KIT	SOUND
Potential	The original all-rounder
808 Boom	Round booming kick, classic metal hats
909 Punch	Clicky driven kick, noisy snare, rides
606 Tight	Small, snappy and crisp
Trap 808	Punch kick, long distorted 808 to tune
SP Dust	12-bit boom bap, crunchy and warm
Electro 80s	Gated snare, big pitch-drop toms
Minimal Tech	Clicks, blips and tight kicks
Industrial	Distorted, metallic and heavy

- Tap a kit to load it and hear a short preview. **PLAY DEMO BEAT** plays a one-bar pattern.
- **RESET PAD KNOBS** puts every pad's tune, length and level back to normal.
- **RECENT KITS** lists the kits you used last. **ADD FOLDER** loads a folder of WAV files: the first 16 (in A-Z order) go on pads 1-16. Name them `01 kick.wav`, `02 808.wav` and so on to choose the order.

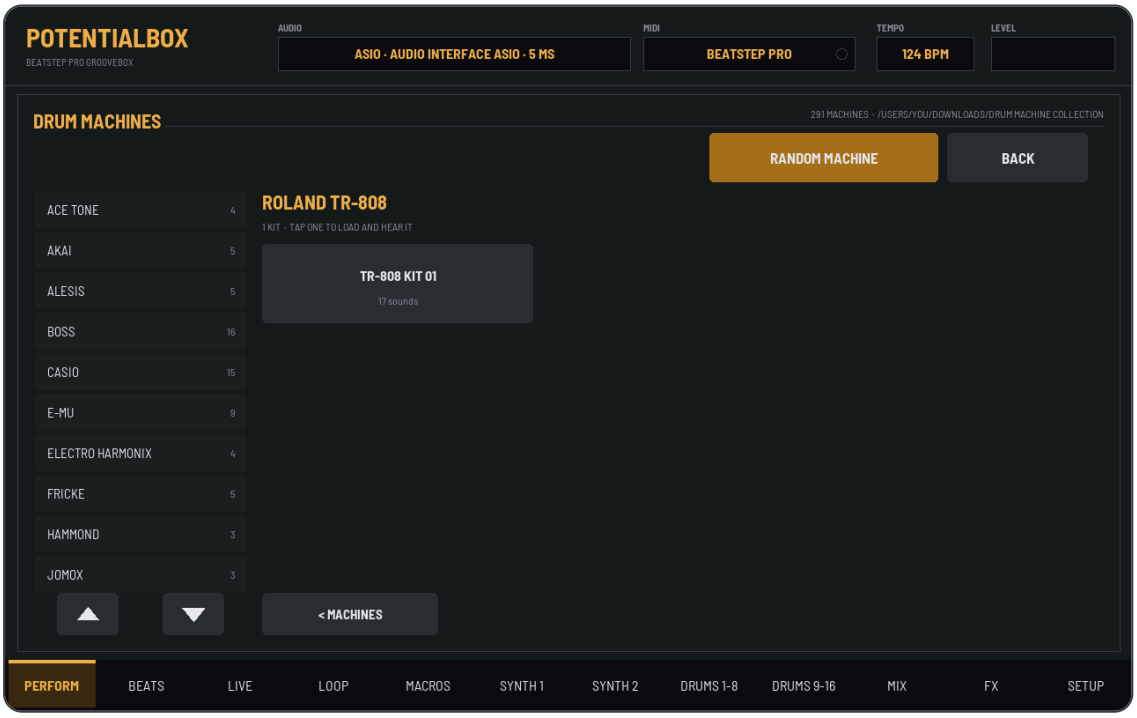
Drum machine library

Point PotentialBox at a drum machine collection (one folder per machine, such as *Roland TR-808*) and tap **DRUM MACHINES**. PotentialBox looks for a folder called **Drum Machine Collection** in your Downloads, Documents, Music or Desktop folder, or use **CHANGE FOLDER**.



Drum machines: brands on the left, machines on the right.

- Pick a **brand**, then a **machine**, then one of its **kits**. Brands with fewer than three machines are under **Other**.
- **RANDOM MACHINE** picks any machine and kit for you.
- Samples are read straight from your folder. Nothing is copied.
- PotentialBox **sorts the sounds onto the right pads by their names** (BD, Kick, SD, Snare, CH, OH, Hat, Tom, Rim, Clap and many more). Spare sounds fill any empty pads.
- With a machine kit loaded, < > step through that machine's kits, then on to the next machine.



The kits inside one machine. Tap one to load and hear it.

Drum knobs

The **Drums 1-8** and **Drums 9-16** tabs each have three pages:

- **MAIN**: a **tune** knob (in semitones) and a **length** knob (shortens the sound) for each pad.
- **MIX**: a **level** and a **pan** knob for each pad.
- **GROOVE**: **chance** and **plays when** for each pad (below).



Drums 1-8, MIX page.

Groove: chance and conditions



Drums 1-8, GROOVE page.

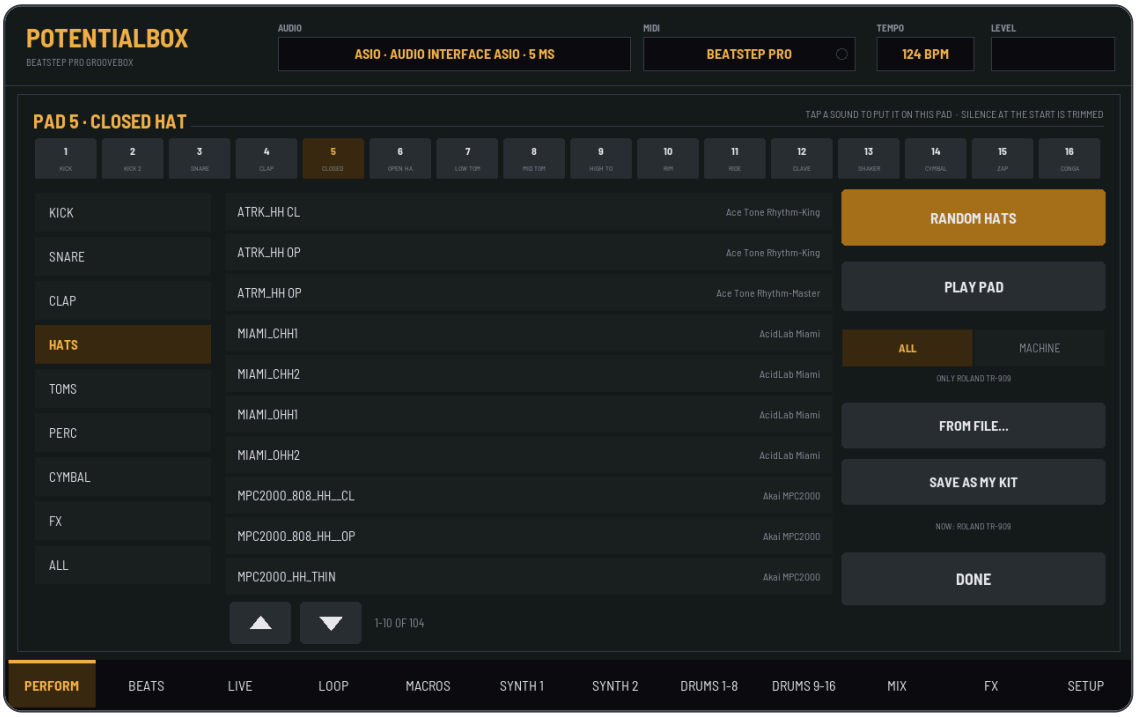
These make a short BeatStep pattern feel longer and more alive, like Elektron's trig conditions.

- **Chance**: at 100% the pad always plays. At 70% about 3 hits in 10 are skipped, at random. Great on hats and percussion.
- **Plays when**: Always, Fill only, Not in fill, 1st of 2, 2nd of 2, 1st of 4, 2nd of 4, 3rd of 4, 4th of 4, 1st of 8, 8th of 8.
- **1st of 4** means "only in the first bar of every four", **4th of 4** only in the fourth: build a crash on bar 1 or a fill on bar 4 from a one-bar pattern. Bars are counted from when you press Play on the BeatStep.
- **Fill only** pads play only while you hold **FILL** (Live tab, Live pad 9, or the Fill pad action). **Not in fill** pads drop out while you hold it.

12 Build your own kit

Put any single drum sound on any pad and save the result as your own kit.

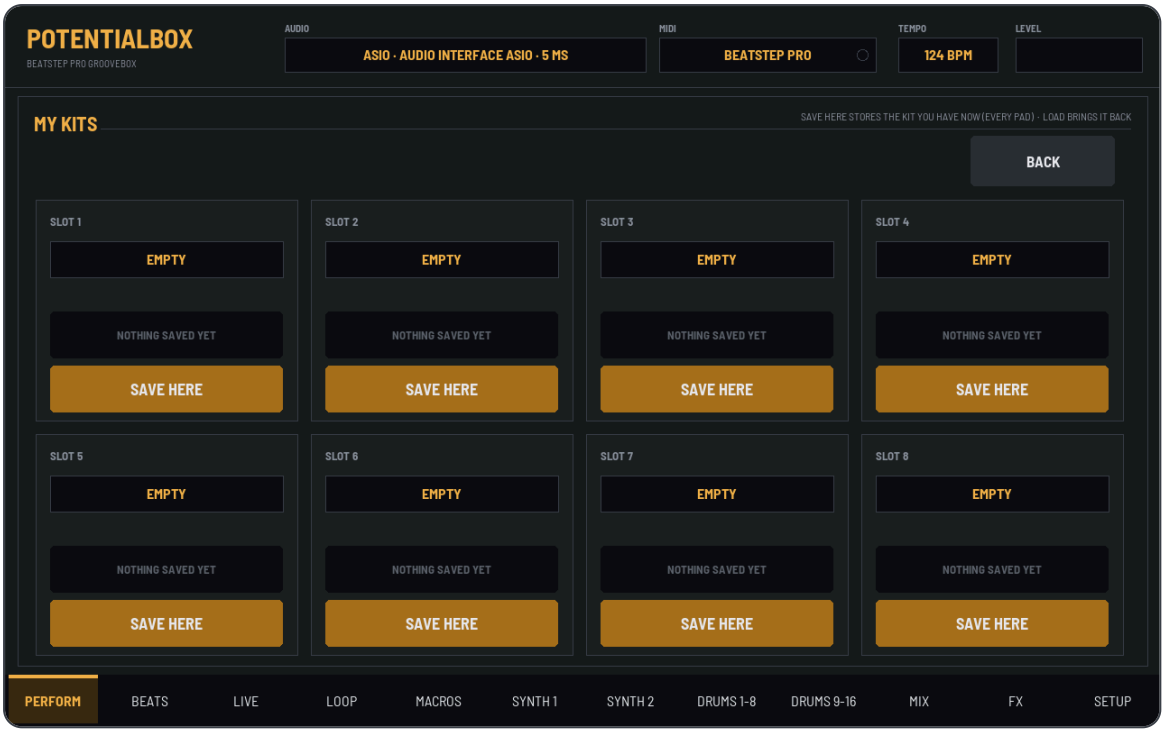
1. Tap a pad on the drum panel. It plays and gets an amber outline.
2. Tap **EDIT PAD**.
3. Pick a type on the left: Kick, Snare, Clap, Hats, Toms, Perc, Cymbal, FX or All. The list shows every matching sound in your whole collection, with the machine it comes from.
4. Tap a sound. It goes on the pad and plays straight away. Keep tapping until you like it.
5. Use the pad strip along the top to move to the next pad.
6. Tap **SAVE AS MY KIT** and choose a slot.



The pad editor.

- **RANDOM** picks a random sound of the chosen type. Great for happy accidents.
- **ALL / MACHINE**: MACHINE limits the list to the drum machine you have loaded.
- **FROM FILE...** takes any WAV file on your PC.
- **Silence at the start of every drum sound is trimmed automatically**, so hits land exactly on the beat.
- An edited kit shows a * after its name and comes back when you restart, even before you save it.

My Kits



My Kits: 8 slots for your own kits.

Open it from the Kit menu or the pad editor. SAVE HERE stores every pad of the current kit; a full slot needs two taps to replace. LOAD brings the kit back.

13 Macros

The **MACROS** tab turns the BeatStep's 16 knobs into performance controls. Each macro moves several settings at once in a musical way: sweeps, throws, risers and drops.



The Macros tab. The top of the page describes the last macro you touched.

> **Macros never overwrite your sound.** They work on top of your settings, so turning a macro back to zero puts everything back exactly where it was. **DJ Filter** rests in the centre. **ZERO ALL MACROS** resets them all at once.

KNOB	MACRO	WHAT IT DOES
1	**S1 Sweep**	Opens Synth 1, adds resonance, throws into delay near the top
2	**S2 Sweep**	Scans Synth 2's wavetable and filter, wobble at the top
3	**Acid**	Synth 1 squelch: resonance, envelope, drive and slide
4	**Wobble**	Synth 2 LFO wobble, deeper and faster
5	**Bright**	Opens up both synths
6	**Pluck**	Both synths short and stabby
7	**Glide**	Both synths slide between notes
8	**Wide**	Detunes both synths into a fat chorus
9	**Dub Throw**	Delay sends up, repeats get longer and darker
10	**Space**	Reverb on everything, bigger room as it goes
11	**DJ Filter**	Centre = off. Left = low-pass, right = high-pass
12	**Build-Up**	Riser: high-pass, reverb and repeats climb, kick drops out at the top
13	**Lo-Fi**	Bit-crush and dirt on the whole mix

KNOB	MACRO	WHAT IT DOES
14	**Drum Smash**	Heavy drum drive
15	**Tighten**	Hats, cymbals and snare get short
16	**Glue**	Master saturation that pulls the mix together

IDEAS

- **Build-Up** then back to zero gives an instant drop: the kick comes back in on the way down.
- Ride **Dub Throw** on the last beat of a phrase, then pull it back.
- **DJ Filter** left for a muffled breakdown, right for a thin, tense one.
- **Acid** plus a little **Glide** on Synth 1 basslines.

14 The Beats tab

The Beats tab makes drum beats in the style of a genre. Tap a genre and you get a beat; tap **NEW BEAT** for another take on it, as often as you like. When you have one you like, **SEND TO BEATSTEP** records it into the BeatStep's drum sequencer, where it carries on playing like a beat you programmed yourself.



The Beats tab: knobs on top, the genres, the beat (tap a dot to change it), and the buttons.

Make a beat

1. Tap a **genre**. With **GENRE KIT: ON** its usual kit loads too, and if the BeatStep is stopped the tempo is set for it.
2. Tap PLAY to hear it. While the BeatStep plays, the beat locks to its clock; when it's stopped, the beat runs on its own tempo (change it with – and +).
3. Tap NEW BEAT for a different take, MUTATE to change just one part, and SIMPLER / BUSIER to thin it out or fill it in.
4. Tap any dot in the grid to remove it, or an empty spot to add a hit.

- In the grid, amber dots are accents, white dots normal hits, small grey dots ghost notes, and dots with a **?** play about half the time, so the beat never repeats exactly.
- **HOLD FOR FILL** plays a fill in the genre's style (a snare roll, a tom run or a break) for as long as you hold it.
- The **mute** buttons silence the kick, snares, hats or percussion while you play.

Send it to the BeatStep

1. On the BeatStep, choose the drum sequencer and an **empty** drum pattern (or clear one). For 32-step genres, set the pattern length to 32. If the genre has swing, set the same swing on the BeatStep (shown next to the tempo).

2. In PotentialBox, tap **SEND TO BEATSTEP** (or press pad 15 on the Beats tab). It shows WAITING FOR REC + PLAY.
3. On the BeatStep, press **REC** and **PLAY**. PotentialBox sends the beat once, from step 1, over MIDI on the drum channel, locked to the BeatStep's clock.
4. When it says SENT, press REC on the BeatStep to stop recording. PotentialBox's own beat stops, and from now on the BeatStep plays the pattern (and you can edit it there like any other).

> While sending, PotentialBox plays the beat itself and ignores the notes the BeatStep plays back, so you don't hear it twice. Probability (?) hits are decided once as they're sent, so the BeatStep gets one exact take.

Genres

GENRE	TEMPO	SWING	STEPS	KIT
Techno	130	straight	16	909 Punch
House	124	54%	16	909 Punch
Deep House	120	58%	16	SP Dust
Minimal	126	55%	16	Minimal Tech
UK Garage	132	64%	16	909 Punch
Drum & Bass	174	straight	32	Potential
Jungle	168	52%	32	SP Dust
Dubstep	140	straight	32	808 Boom
Grime	140	straight	16	Electro 80s
UK Drill	142	straight	32	Trap 808
Trap	140	straight	32	Trap 808
Hip Hop	92	58%	16	808 Boom
Boom Bap	88	63%	16	SP Dust
Lo-fi	78	66%	16	SP Dust
Reggae One Drop	76	60%	16	Potential
Steppers	76	58%	16	Potential
Dancehall	100	52%	16	808 Boom
Reggaeton	95	straight	16	Trap 808
Afrobeats	108	56%	16	Potential
Breakbeat	130	52%	16	Electro 80s
Electro	125	straight	16	Electro 80s

GENRE	TEMPO	SWING	STEPS	KIT
Disco	118	52%	16	909 Punch
Funk	100	56%	16	SP Dust
Footwork	160	straight	32	808 Boom

Knobs

KNOB	CONTROL	GROUP
1	Busy	Beat feel
2	Variation	Beat feel
3	Swing	Beat feel
4	Humanize	Beat feel
5	Drums volume	Drum bus
6	Drum filter	Drum bus
7	Drum drive	Drum bus
8	Reverb: Drums	Drum bus
9	Pad 1 level	Pad levels
10	Pad 2 level	Pad levels
11	Pad 3 level	Pad levels
12	Pad 4 level	Pad levels
13	Pad 5 level	Pad levels
14	Pad 6 level	Pad levels
15	Pad 13 level	Pad levels
16	Pad 1 tune	Kick

Busy makes the ? hits more or less likely and thins out the extras; **Variation** adds the odd extra hit (a ghost snare, a hat roll) now and then; **Swing** at "Genre" uses the genre's own swing; **Humanize** loosens the timing and velocity a little.

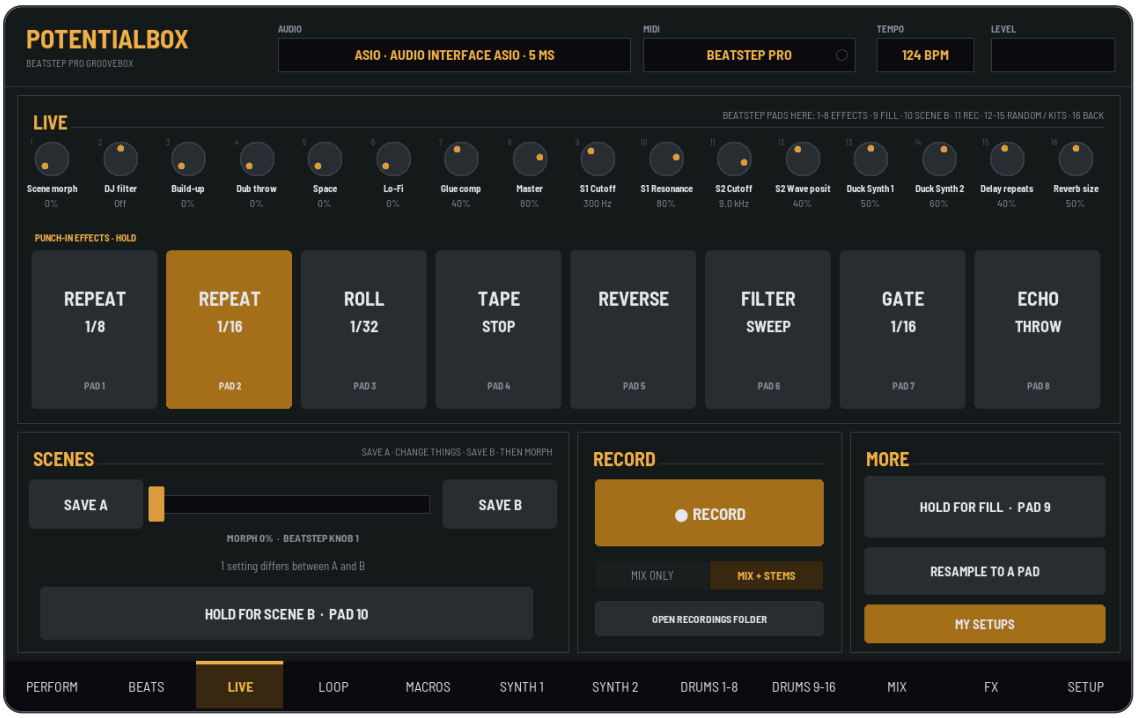
The BeatStep's pads on the Beats tab

PAD	BEATS PAD
1	Play / stop
2	New beat
3	Previous genre

PAD	BEATS PAD
4	Next genre
5	Hold: Fill
6	Mutate
7	Simpler
8	Busier
9	Mute kick
10	Mute snare
11	Mute hats
12	Mute perc
13	Tempo down
14	Tempo up
15	Send to BeatStep
16	Back to last tab

15 The Live tab

The Live tab is for playing: hold-to-play effects, scenes, fill, recording and resampling, all one tap away.



The Live tab.

Knobs

The 16 small knobs are the BeatStep's knobs on this tab: the controls you reach for most when performing.

KNOB	CONTROL	GROUP
1	Scene morph	Scenes
2	DJ filter	Mix and macros
3	Build-Up	Mix and macros
4	Dub Throw	Mix and macros
5	Space	Mix and macros
6	Lo-Fi	Mix and macros
7	Glue comp	Mix and macros
8	Master	Mix and macros
9	S1 Cutoff	Synths
10	S1 Resonance	Synths
11	S2 Cutoff	Synths

KNOB	CONTROL	GROUP
12	S2 Wave position	Synths
13	Duck Synth 1	Duck
14	Duck Synth 2	Duck
15	Delay repeats	Effects
16	Reverb size	Effects

Punch-in effects

Hold a pad to play an effect over the whole mix; let go and the music carries on as before.

PAD	EFFECT	WHAT IT DOES
1	**Repeat 1/8**	Loops the eighth note you pressed on, over and over (beat repeat).
2	**Repeat 1/16**	The same with a sixteenth: a stutter.
3	**Roll 1/32**	A fast drum-roll buzz.
4	**Tape stop**	The music slows down to a stop over one beat, like a turntable switched off.
5	**Reverse**	Plays the music backwards from the moment you pressed.
6	**Filter sweep**	A resonant low-pass closes over two beats; it opens again when you let go.
7	**Gate 1/16**	Chops the sound on and off in sixteenths (trance gate).
8	**Echo throw**	Sends the mix into a long dub echo. The echoes keep ringing after you let go.

Only one of pads 1-5 works at a time (the last one you pressed); pads 6-8 can be held on top.

Scenes

1. Get a sound you like and tap SAVE A.
2. Change anything: filters, effects, levels, drum tuning, macros. Then tap SAVE B.
3. Drag the **morph** bar (or turn BeatStep knob 1) to blend from A to B. Only the settings that differ move.
4. **HOLD FOR SCENE B** (Live pad 10) jumps to B while held and goes back when you let go: an instant drop.

> Morphing sets the knobs it moves. If you change one of those knobs by hand, moving the morph again takes it back to the A to B blend. Save the scene again to keep your change.

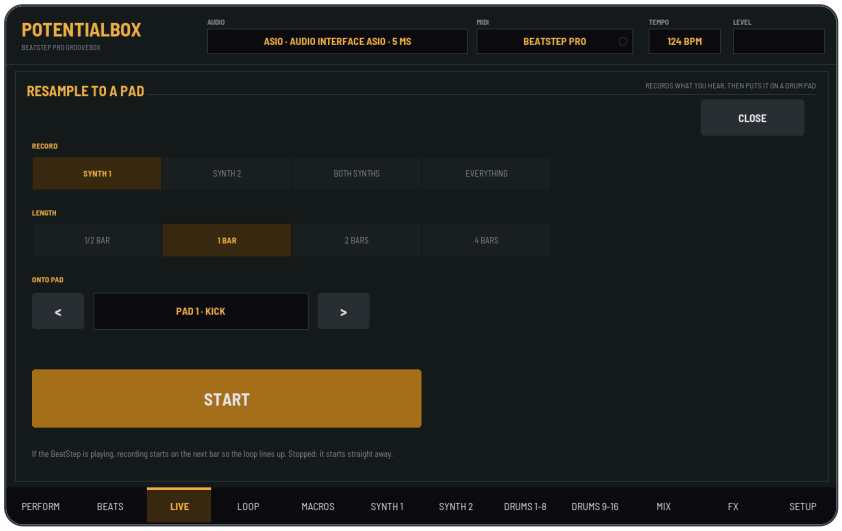
Fill

HOLD FOR FILL (Live pad 9) plays the drum pads set to **Fill only** and silences those set to **Not in fill** (chapter 11).

Record

- Tap RECORD (or Live pad 11) to start, and again to stop. The header shows the time while recording.
- Recordings go into **Music > PotentialBox Recordings**, one folder per take, as 24-bit WAV files.
- **MIX + STEMS** also saves Drums, Synth 1, Synth 2 and FX returns as separate files that line up, ready to remix in any DAW or on an MPC. **MIX ONLY** saves just the master.
- The stems are taken before the master effects (DJ filter, EQ, glue, punch-in effects); the master file has everything.

Resample to a pad

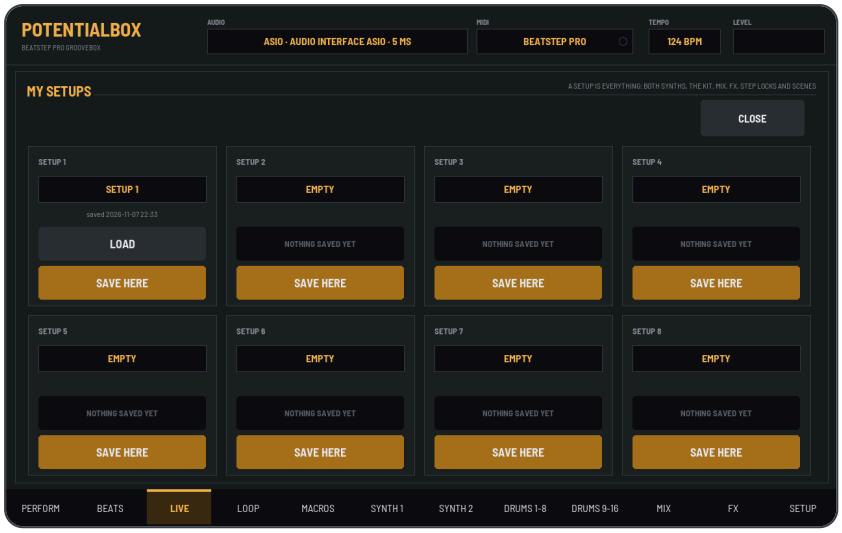


Resample: choose what to record, how long, and which pad.

1. Tap RESAMPLE TO A PAD.
2. Choose what to record (Synth 1, Synth 2, both synths or everything), how long (half a bar to 4 bars) and the pad.
3. Tap START. If the BeatStep is playing, it waits for the next bar so the loop lines up.
4. When it's done, the sound is on the pad (as part of your current kit) and plays once. Save the kit in My Kits to keep it.

My setups

A setup stores everything at once: both synth sounds, the drum kit, every knob, step locks and scenes. Tap MY SETUPS, then **SAVE HERE** on a slot (tap twice to replace one). **LOAD** brings the whole setup back, for example one per song.



My setups: 8 slots for complete setups.

The BeatStep's pads on the Live tab

While the Live tab is open, the 16 pads that normally run pad actions play the Live pads instead, so you can perform without looking at the screen:

PAD	LIVE PAD
1	Hold: Repeat 1/8
2	Hold: Repeat 1/16
3	Hold: Roll 1/32
4	Hold: Tape stop
5	Hold: Reverse
6	Hold: Filter sweep
7	Hold: Gate 1/16
8	Hold: Echo throw
9	Hold: Fill
10	Hold: Scene B
11	Record on/off
12	Random Synth 1
13	Random Synth 2
14	Random Drum Kit
15	Next Kit
16	Back to last tab

Pad 16 goes back to the tab you came from, where the pads run their normal actions again.

16 The Loop tab

The Loop tab is a 5-track looper, inspired by the BOSS RC-505: record a phrase, it plays round and round, add more on top or on other tracks, and build a whole song live. It records the inputs of your audio interface (mic, guitar, keys, other gear) and PotentialBox's own synths and drums.



The Loop tab: five tracks, the 16 knobs on top, settings on the right.

First loop in four taps

1. Tap INPUT OFF at the top right and pick the input your mic or instrument is plugged into (for example **IN 1**, or **IN 1+2 STEREO**). This needs your interface's **ASIO** driver chosen in SETUP.
2. Press **pad 1** (or tap REC on track 1) and play.
3. Press pad 1 again: the loop plays. Pressing it again starts an **overdub**: what you play now is added on top. Once more goes back to playing.
4. Pad 2 records track 2, and so on. **Hold** a track's stop pad (9-13) for a second to clear it.

Each track

- The big button shows what a tap does next: **REC**, **PLAY** or **DUB**. Colours: **red** recording, **green** playing, **yellow** overdubbing. WAITING and FINISHING flash while it waits for the bar.
- The bar under the name shows where the loop is; under it is the length.
- **FROM** chooses what this track records: Input, PotentialBox, Both. "PotentialBox" records the synths and drums (and their effects), without the loops, so nothing doubles up.
- **FX** picks the track's effect (Low-pass, High-pass, Lo-fi, Echo, Tremolo). Its amount is the track's FX knob (knobs 9-13).

- **REVERSE** plays the track backwards. **SPEED** switches between 1/2 (an octave down, like slowing a tape), 1x and 2x. Overdubbing works at 1x.

Timing

- **SYNC TO BEATSTEP** (on): while the BeatStep plays, every loop's length snaps to whole bars at its tempo, and the loops follow its Play and Stop. Pressing Play on the BeatStep starts all loops from the top, in time with bar 1.
- With the BeatStep stopped, the **first loop** is as long as you play it, and later loops snap to whole repeats of it.
- **START: WHEN I TAP** starts recording the moment you press; the end still snaps, so you never wait. **START: ON THE BAR** waits for the next bar instead. Pressing a little late still counts as on the bar.
- **LENGTH**: Free, 1 bar, 2 bars, 4 bars, 8 bars. With a length set, recording stops by itself after that many bars and the loop starts playing: just press once.
- Your interface's delay is measured and made up for, so what you record lands in time with what you hear.

Knobs

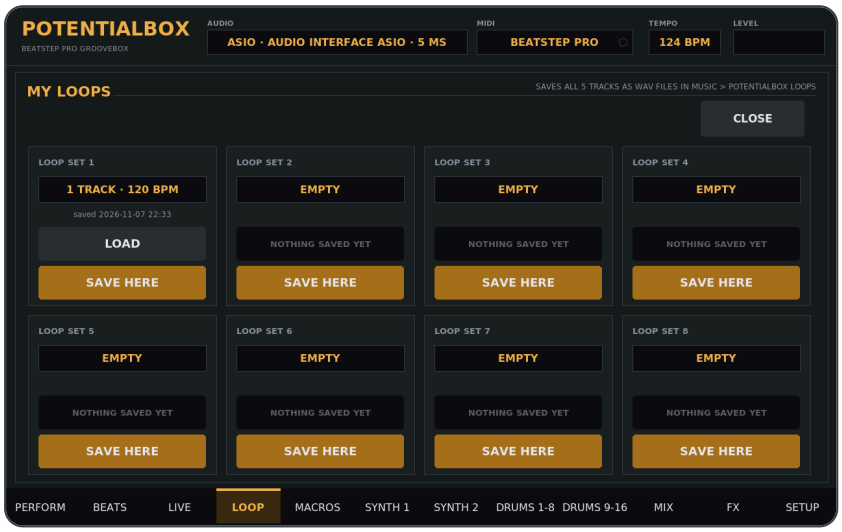
KNOB	CONTROL	WHAT IT DOES
1	Track 1 level	Track 1's level.
2	Track 2 level	Track 2's level.
3	Track 3 level	Track 3's level.
4	Track 4 level	Track 4's level.
5	Track 5 level	Track 5's level.
6	Input level	How loud the interface input is recorded. Watch the meter under the INPUT button.
7	Hear input	How loud you hear the input live. Turn it down if your interface already lets you hear it.
8	Looper level	Level of all the loops together.
9	Track 1 FX	Amount of track 1's effect (0% = off).
10	Track 2 FX	Amount of track 2's effect (0% = off).
11	Track 3 FX	Amount of track 3's effect (0% = off).
12	Track 4 FX	Amount of track 4's effect (0% = off).
13	Track 5 FX	Amount of track 5's effect (0% = off).
14	Overdub keep	How much of the old loop stays each time round while overdubbing. Below 100% older layers slowly fade.
15	Loop filter	DJ filter on all the loops: left low-pass, right high-pass.
16	Duck loops	The loops dip when the kick plays, using the side-chain settings on the Mix tab.

Undo, stop and clear

- **UNDO / REDO** (pad 6) takes back the last overdub on the last track you used; press again to bring it back. Undoing a first recording empties the track (press again to get it back).
- **ALL STOP** (pad 7) stops every track; pressed again it is **ALL PLAY**.
- **CLEAR ALL** empties every track (tap twice). Pad 8 records on the next empty track.

My loops

MY LOOPS saves all five tracks in one of 8 slots, as WAV files in **Music > PotentialBox Loops** (one folder per loop set), ready to drag into a DAW or onto an MPC. LOAD brings them back stopped: press ALL PLAY or Play on the BeatStep. The tempo they were saved at is shown, so you can set the BeatStep to match.



My loops: 8 loop sets.

The BeatStep's pads on the Loop tab

While the Loop tab is open, the 16 pads play the looper. Each column is one track: the bottom pad records and plays, the pad above it stops (hold it to clear).

PAD	LOOP PAD
1	Track 1 rec / play / dub
2	Track 2 rec / play / dub
3	Track 3 rec / play / dub
4	Track 4 rec / play / dub
5	Track 5 rec / play / dub
6	Undo / redo
7	All stop / play
8	Record next empty track
9	Track 1 stop (hold: clear)
10	Track 2 stop (hold: clear)

PAD	LOOP PAD
11	Track 3 stop (hold: clear)
12	Track 4 stop (hold: clear)
13	Track 5 stop (hold: clear)
14	Reverse last track
15	Speed last track
16	Back to last tab

> **Recording:** the RECORD button on the Live tab also saves a **Looper** stem with everything the loops played. **With a mic**, use headphones so the speakers don't feed back into it.

17 Step locks

A step lock gives a knob a different value on one step of the pattern: a filter that opens on step 4, a pitch drop on the last step, a snare with a longer tail on step 13. The same idea as Elektron's parameter locks.



Step locks on Synth 1: step 3 is selected and the cutoff knob is locked there.

1. Open the page with the knob you want, then tap **STEP LOCKS** at the top right.
2. The panels at the bottom turn into **16 step buttons**. Tap a step: it turns amber.
3. Turn knobs (on screen or on the BeatStep). Now they only set that step, and the knob shows **LOCK**.
4. Tap another step to lock that one, or tap the same step again to go back to normal knobs. **DONE** closes the panel.

- Locks play while the BeatStep is playing. They follow its clock: step 1 is the first step after you press Play, and the 16 steps repeat over and over (for patterns longer than 16 steps, the locks repeat every 16).
- Every knob except the macros, scene morph and master volume can be locked: synths, drums (including chance and pan), mix and effects.
- A knob with locks shows a small amber **L**. Steps with locks show how many on their button.
- **CLEAR STEP**, **CLEAR PAGE** (the knobs on this page) and **CLEAR ALL** remove locks. The INIT buttons on the synth pages and RESET ALL TO INIT also clear the locks they cover.
- Locks are saved with your settings and in My Setups. They are not part of My Sounds.

18 Mix and FX



The Mix tab.

Mix tab

KNOB	CONTROL	GROUP
1	Synth 1 volume	Levels
2	Synth 2 volume	Levels
3	Drums volume	Levels
4	Master	Levels
5	Drum filter	Bus & DJ filter
6	Drum drive	Bus & DJ filter
7	Master drive	Bus & DJ filter
8	DJ filter	Bus & DJ filter
9	Duck Synth 1	Side-chain duck
10	Duck Synth 2	Side-chain duck
11	Duck FX returns	Side-chain duck
12	Duck from	Side-chain duck
13	Duck release	Side-chain duck
14	Glue comp	Glue & width
15	Glue speed	Glue & width
16	Stereo width	Glue & width

RESET ALL TO INIT at the top of the Mix tab resets every knob in the app (both synths, all pads, mix, effects and macros), and clears step locks and scenes. Tap it twice. Saved sounds, kits, setups and the audio setup are kept.

SIDE-CHAIN DUCK

Ducking makes the synths dip each time the kick plays, so the kick punches through: the "pumping" sound of house and techno.

- **Duck Synth 1**, **Duck Synth 2** and **Duck FX returns** set how far each one dips. 0% is off.
- **Duck from** chooses the pad that triggers it (pad 1, the kick, out of the box), or **Every beat** to pump in time without any kick.
- **Duck release** sets how quickly the sound comes back: short for a tight bounce, long for big pumping.

GLUE AND WIDTH

- **Glue comp** is a one-knob bus compressor on the whole mix: more means more squeeze, with the level made up for you. **Glue speed**: Fast, Medium, Slow.
- **Stereo width**: 100% is normal, 0% is mono, up to 200% is extra wide.



The Mix tab, EQ page.

EQ page

KNOB	CONTROL	GROUP
1	Synth 1 Low	Synth 1 EQ
2	Synth 1 Mid	Synth 1 EQ
3	Synth 1 High	Synth 1 EQ
4	Synth 2 Low	Synth 2 EQ
5	Synth 2 Mid	Synth 2 EQ
6	Synth 2 High	Synth 2 EQ
7	Low EQ freq	EQ frequencies
8	High EQ freq	EQ frequencies

KNOB	CONTROL	GROUP
9	Drums Low	Drums EQ
10	Drums Mid	Drums EQ
11	Drums High	Drums EQ
12	Master Low	Master EQ
13	Master Mid	Master EQ
14	Master High	Master EQ
15	S1 Pan	Synth pan
16	S2 Pan	Synth pan

Each EQ has a low shelf, a mid band around 1 kHz and a high shelf, from -12 to +12 dB. **Low EQ freq** and **High EQ freq** set where the low and high shelves start, for all four EQs.



The FX tab.

FX tab

KNOB	CONTROL	GROUP
1	Delay: Synth 1	Delay sends
2	Delay: Synth 2	Delay sends
3	Delay: Drums	Delay sends
4	Delay time	Delay
5	Delay repeats	Delay
6	Delay tone	Delay
7	DJ filter	DJ filter & lo-fi
8	Lo-fi crush	DJ filter & lo-fi

KNOB	CONTROL	GROUP
9	Reverb: Synth 1	Reverb sends
10	Reverb: Synth 2	Reverb sends
11	Reverb: Drums	Reverb sends
12	Reverb size	Reverb
13	Reverb tone	Reverb
14	Reverb level	Reverb
15	Drum filter	Bus
16	Master drive	Bus

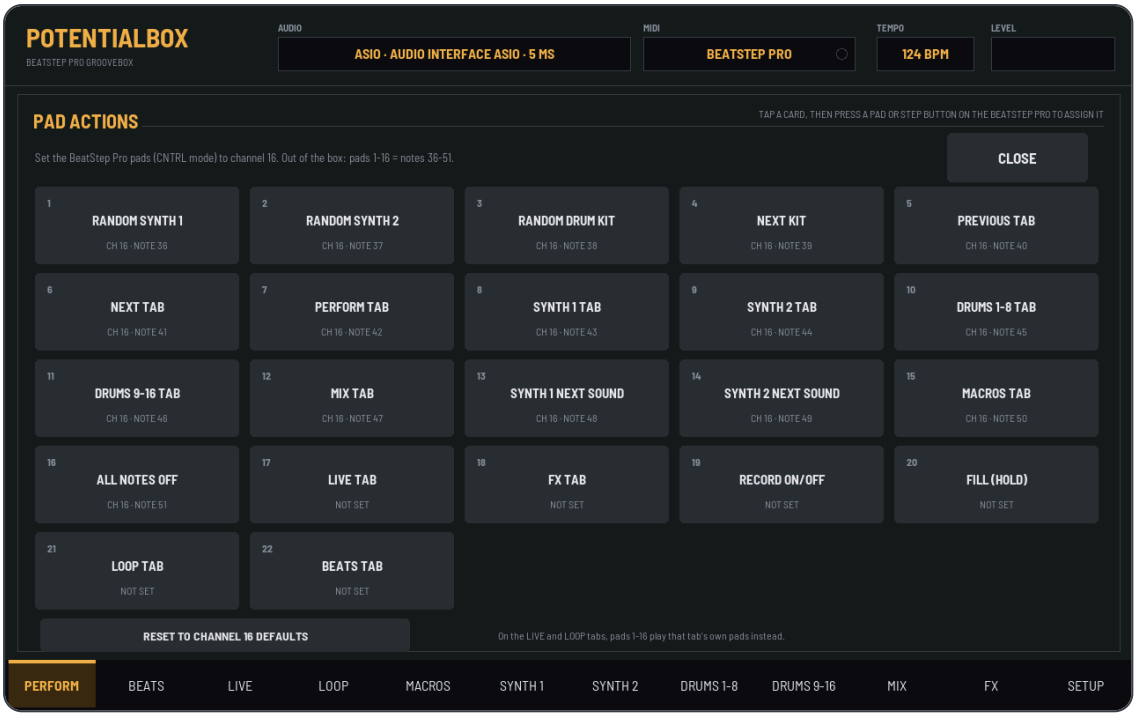
- **Delay**: a ping-pong delay synced to the BeatStep's tempo (1/16, 1/8T, 1/8, 3/16, 1/4, 3/8). Each instrument has its own send.
- **Reverb**: a smooth room-to-hall reverb with a send for each instrument. Size sets how long it rings.
- **DJ filter**: centre is off, left is low-pass, right is high-pass. It works on the whole mix.
- **Lo-fi crush**: lowers the sample rate and bit depth of the whole mix.

19 Pad actions

With the BeatStep Pro's Control Mode pads on **MIDI channel 16** (chapter 5), each pad runs a PotentialBox action. Notes on channel 16 never play the synths.

PAD	NOTE	ACTION
1	36	**Random Synth 1**
2	37	**Random Synth 2**
3	38	**Random Drum Kit**
4	39	**Next Kit**
5	40	**Previous Tab**
6	41	**Next Tab**
7	42	**Perform Tab**
8	43	**Synth 1 Tab**
9	44	**Synth 2 Tab**
10	45	**Drums 1-8 Tab**
11	46	**Drums 9-16 Tab**
12	47	**Mix Tab**
13	48	**Synth 1 Next Sound**
14	49	**Synth 2 Next Sound**
15	50	**Macros Tab**
16	51	**All Notes Off**
17	not set	**Live Tab**
18	not set	**FX Tab**
19	not set	**Record On/Off**
20	not set	**Fill (hold)**
21	not set	**Loop Tab**
22	not set	**Beats Tab**

Changing them



SETUP > PAD ACTIONS.

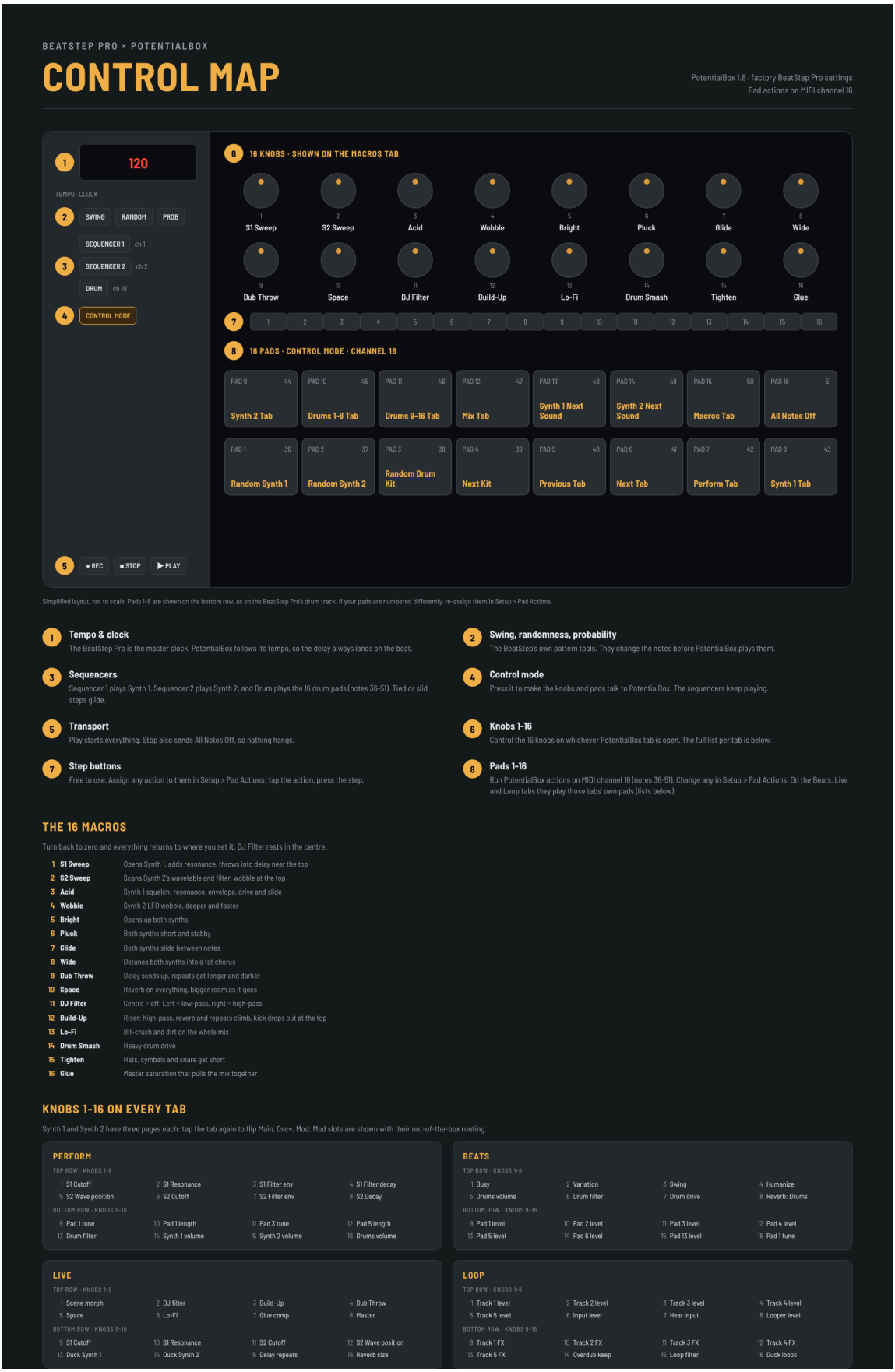
1. Open **SETUP** and tap **PAD ACTIONS**.
2. Tap the action you want to change. It turns amber.
3. Press the pad or step button on the BeatStep Pro you want to use for it. Notes and CCs both work, on any channel.

Each card flashes when its action runs. **RESET TO CHANNEL 16 DEFAULTS** puts the table above back.

Actions 17-20 have no pad out of the box: tap the card and press a spare pad or step button to use them. **Fill (hold)** plays fills for as long as you hold it. On the **Live tab** the pads play the Live pads instead (chapter 15), on the **Loop tab** the Loop pads (chapter 16) and on the **Beats tab** the Beats pads (chapter 14).

20 Control map

Everything on the BeatStep Pro at a glance, with the 16 knobs shown as they work on the Macros tab.



Control map, continued: the 16 knobs on every tab and page, and the Live tab pads.

MACROS

TOP ROW · KNOBS 1–8

1 S1 Sweep	2 S2 Sweep	3 Acid	4 Wobble
5 Bright	6 Pluck	7 Glide	8 Wide

BOTTOM ROW · KNOBS 9–16

9 Dub Throw	10 Space	11 DJ Filter	12 Build-Up
13 Lo-Fi	14 Drum Smash	15 Tighten	16 Glue

SYNTH 1 · MAIN

TOP ROW · KNOBS 1–8

1 Cutoff	2 Resonance	3 Filter env	4 Filter decay
5 Saw > Square	6 Pulse width	7 Detune	8 Sub osc

BOTTOM ROW · KNOBS 9–16

9 Attack	10 Decay	11 Sustain	12 Release
13 Glide (slides)	14 Octave	15 Drive	16 Synth 1 volume

SYNTH 1 · OSC+

TOP ROW · KNOBS 1–8

1 Osc 2 pitch	2 FM (osc 2 > 1)	3 Ring mod	4 Hard sync
5 Noise	6 Chord	7 Chord voicing	8 Pan

BOTTOM ROW · KNOBS 9–16

9 Filter type	10 Key tracking	11 Velocity > volume	12 Analog drift
13 Pitch env	14 Pitch env time	15 Glide mode	16 Synth 1 volume

SYNTH 1 · MOD

TOP ROW · KNOBS 1–8

1 LFO 1 shape	2 LFO 1 speed	3 LFO 1 sync	4 LFO 2 shape
5 LFO 2 speed	6 LFO 2 sync	7 Mod env attack	8 Mod env decay

BOTTOM ROW · KNOBS 9–16

9 LFO 1 > Pitch	10 LFO 1 > Cutoff	11 LFO 2 > Pulse width	12 Mod env > Cutoff
13 Velocity > Cutoff	14 LFO 2 > Pan	15 Mod env > Pitch	16 LFO 2 > Volume

SYNTH 1 · ARP+

TOP ROW · KNOBS 1–8

1 Arp	2 Arp rate	3 Arp octaves	4 Arp gate
5 Arp swing	6 Arp hold	7 Arp chance	8 Arp accent

BOTTOM ROW · KNOBS 9–16

9 Duck Synth 1	10 Synth 1 Low	11 Synth 1 Mid	12 Synth 1 High
13 Pan	14 Delay: Synth 1	15 Reverb: Synth 1	16 Synth 1 volume

SYNTH 2 · MAIN

TOP ROW · KNOBS 1–8

1 Wave position	2 Env → Position	3 LFO speed	4 LFO → Position
5 Cutoff	6 Resonance	7 Filter env	8 Wavetable

BOTTOM ROW · KNOBS 9–16

9 Attack	10 Decay	11 Sustain	12 Release
13 Glide (slides)	14 Octave	15 Unison	16 Synth 2 volume

SYNTH 2 · OSC+

TOP ROW · KNOBS 1–8

1 Chord	2 Chord voicing	3 Sub osc	4 Noise
5 Pan	6 Analog drift	7 Pitch env	8 Pitch env time

BOTTOM ROW · KNOBS 9–16

9 Filter type	10 Key tracking	11 Velocity > volume	12 Glide mode
13 Wavetable	14 Wave position	15 Cutoff	16 Synth 2 volume

SYNTH 2 · MOD

TOP ROW · KNOBS 1–8

1 LFO 1 shape	2 LFO 1 speed	3 LFO 1 sync	4 LFO 2 shape
5 LFO 2 speed	6 LFO 2 sync	7 Mod env attack	8 Mod env decay

BOTTOM ROW · KNOBS 9–16

9 LFO 1 > Pitch	10 LFO 1 > Cutoff	11 LFO 2 > Wave position	12 Mod env > Wave position
13 Velocity > Cutoff	14 LFO 2 > Pan	15 Mod env > Pitch	16 LFO 2 > Volume

SYNTH 2 · ARP+

TOP ROW · KNOBS 1–8

1 Arp	2 Arp rate	3 Arp octaves	4 Arp gate
5 Arp swing	6 Arp hold	7 Arp chance	8 Arp accent

BOTTOM ROW · KNOBS 9–16

9 Duck Synth 2	10 Synth 2 Low	11 Synth 2 Mid	12 Synth 2 High
13 Pan	14 Delay: Synth 2	15 Reverb: Synth 2	16 Synth 2 volume

DRUMS 1–8 · MAIN

TOP ROW · KNOBS 1–8

1 Pad 1 tune	2 Pad 2 tune	3 Pad 3 tune	4 Pad 4 tune
5 Pad 5 tune	6 Pad 6 tune	7 Pad 7 tune	8 Pad 8 tune

BOTTOM ROW · KNOBS 9–16

9 Pad 1 length	10 Pad 2 length	11 Pad 3 length	12 Pad 4 length
13 Pad 5 length	14 Pad 6 length	15 Pad 7 length	16 Pad 8 length

DRUMS 1–8 · MIX

TOP ROW · KNOBS 1–8

1 Pad 1 level	2 Pad 2 level	3 Pad 3 level	4 Pad 4 level
5 Pad 5 level	6 Pad 6 level	7 Pad 7 level	8 Pad 8 level

BOTTOM ROW · KNOBS 9–16

9 Pad 1 pan	10 Pad 2 pan	11 Pad 3 pan	12 Pad 4 pan
13 Pad 5 pan	14 Pad 6 pan	15 Pad 7 pan	16 Pad 8 pan

DRUMS 1–8 · GROOVE

TOP ROW · KNOBS 1–8

1 Pad 1 chance	2 Pad 2 chance	3 Pad 3 chance	4 Pad 4 chance
5 Pad 5 chance	6 Pad 6 chance	7 Pad 7 chance	8 Pad 8 chance

BOTTOM ROW · KNOBS 9–16

9 Pad 1 plays	10 Pad 2 plays	11 Pad 3 plays	12 Pad 4 plays
13 Pad 5 plays	14 Pad 6 plays	15 Pad 7 plays	16 Pad 8 plays

DRUMS 9–16 · MAIN

TOP ROW · KNOBS 1–8

1 Pad 9 tune	2 Pad 10 tune	3 Pad 11 tune	4 Pad 12 tune
5 Pad 13 tune	6 Pad 14 tune	7 Pad 15 tune	8 Pad 16 tune

BOTTOM ROW · KNOBS 9–16

9 Pad 9 length	10 Pad 10 length	11 Pad 11 length	12 Pad 12 length
13 Pad 13 length	14 Pad 14 length	15 Pad 15 length	16 Pad 16 length

DRUMS 9–16 · MIX

TOP ROW · KNOBS 1–8

1 Pad 9 level	2 Pad 10 level	3 Pad 11 level	4 Pad 12 level
5 Pad 13 level	6 Pad 14 level	7 Pad 15 level	8 Pad 16 level

BOTTOM ROW · KNOBS 9–16

9 Pad 9 pan	10 Pad 10 pan	11 Pad 11 pan	12 Pad 12 pan
13 Pad 13 pan	14 Pad 14 pan	15 Pad 15 pan	16 Pad 16 pan

DRUMS 9–16 · GROOVE

TOP ROW · KNOBS 1–8

1 Pad 9 chance	2 Pad 10 chance	3 Pad 11 chance	4 Pad 12 chance
5 Pad 13 chance	6 Pad 14 chance	7 Pad 15 chance	8 Pad 16 chance

BOTTOM ROW · KNOBS 9–16

9 Pad 9 plays	10 Pad 10 plays	11 Pad 11 plays	12 Pad 12 plays
13 Pad 13 plays	14 Pad 14 plays	15 Pad 15 plays	16 Pad 16 plays

MIX · MAIN

TOP ROW · KNOBS 1–8

1 Synth 1 volume	2 Synth 2 volume	3 Drums volume	4 Master
5 Drum filter	6 Drum drive	7 Master drive	8 DJ filter

BOTTOM ROW · KNOBS 9–16

9 Duck Synth 1	10 Duck Synth 2	11 Duck FX returns	12 Duck from
13 Duck release	14 Glue comp	15 Glue speed	16 Stereo width

MIX · EQ

TOP ROW · KNOBS 1–8

1 Synth 1 Low	2 Synth 1 Mid	3 Synth 1 High	4 Synth 2 Low
5 Synth 2 Mid	6 Synth 2 High	7 Low EQ freq	8 High EQ freq

BOTTOM ROW · KNOBS 9–16

9 Drums Low	10 Drums Mid	11 Drums High	12 Master Low
13 Master Mid	14 Master High	15 S1 Pan	16 S2 Pan

FX

TOP ROW · KNOBS 1–8

1 Delay: Synth 1	2 Delay: Synth 2	3 Delay: Drums	4 Delay time
5 Delay repeats	6 Delay tone	7 DJ filter	8 Lo-fi crush

BOTTOM ROW · KNOBS 9–16

9 Reverb: Synth 1	10 Reverb: Synth 2	11 Reverb: Drums	12 Reverb size
13 Reverb tone	14 Reverb level	15 Drum filter	16 Master drive

PADS 1–16 ON THE BEATS TAB

1 Play / stop	2 New beat	3 Previous genre	4 Next genre
5 Hold Fill	6 Mutate	7 Simpler	8 Busier
9 Mute kick	10 Mute snare	11 Mute hats	12 Mute perc
13 Tempo down	14 Tempo up	15 Send to BeatStep	16 Back to last tab

PADS 1–16 ON THE LIVE TAB

1 Hold Repeat 1/8	2 Hold Repeat 1/16	3 Hold Roll 1/32	4 Hold Tape stop
5 Hold Reverse	6 Hold Filter sweep	7 Hold Gate 1/16	8 Hold Echo throw
9 Hold Fill	10 Hold Scene B	11 Record on/off	12 Random Synth 1
13 Random Synth 2	14 Random Drum Kit	15 Next Kit	16 Back to last tab

PADS 1–16 ON THE LOOP TAB

1 Track 1 rec / play / dub	2 Track 2 rec / play / dub
3 Track 3 rec / play / dub	4 Track 4 rec / play / dub
5 Track 5 rec / play / dub	6 Undo / redo
7 All stop / play	8 Record next empty track
9 Track 1 stop (hold: clear)	10 Track 2 stop (hold: clear)
11 Track 3 stop (hold: clear)	12 Track 4 stop (hold: clear)
13 Track 5 stop (hold: clear)	14 Reverse last track
15 Speed last track	16 Back to last tab

21 Your files

PotentialBox saves everything by itself, a moment after each change and when you close it.

WHAT	WHERE
The app	`%LOCALAPPDATA%\PotentialBox`
Settings (knobs, setup, pad actions, current kit)	`%APPDATA%\PotentialBox\state.json`
My Sounds	`%APPDATA%\PotentialBox\my-sounds.json`
My Kits	`%APPDATA%\PotentialBox\my-kits.json`
My Setups	`%APPDATA%\PotentialBox\my-setups.json`
Resampled sounds	`%APPDATA%\PotentialBox\Resampled`
Recordings	`Music\PotentialBox Recordings` (one folder per take)
Current beat	saved with your settings and in My Setups
My Loops	`Music\PotentialBox Loops` (one folder per loop set)
Log (what went wrong, if anything)	`%APPDATA%\PotentialBox\log.txt`

To open one of these folders, paste the path (for example `%APPDATA%\PotentialBox`) into the File Explorer address bar and press Enter.

> **Back up** your sounds and kits by copying `my-sounds.json` and `my-kits.json` somewhere safe. My Kits point to your sample files, so keep your sample folders where they are.

22 Troubleshooting

THE HEADER SAYS AUDIO ERROR

Open SETUP and tap another output. If your interface's ASIO driver won't open, close any DAW, FL Studio or Voicemeeter that might be using it, then tap it again. WASAPI always works as a fallback.

I HEAR CRACKLES

Tap a bigger BUFFER in SETUP (256 or 512). Closing heavy apps also helps.

MIDI SAYS NO BEATSTEP FOUND

Check the USB cable, wait a few seconds after plugging in, then tap RESCAN MIDI in SETUP. Some apps (such as a DAW) can hold the BeatStep's MIDI port open; close them and rescan.

THE SEQUENCERS PLAY BUT THE KNOBS DO NOTHING

Press CONTROL MODE on the BeatStep Pro. Still nothing? SETUP > LEARN KNOBS.

STEP LOCKS, ARP TIMING OR "1ST OF 4" DON'T FOLLOW THE BEATSTEP

They follow the BeatStep's MIDI clock. Press Play on the BeatStep (not just the sequencer pads) and check the TEMPO in the header shows its tempo. Stop and Play again to reset the step and bar count.

SEND TO BEATSTEP SAYS NO MIDI OUT WAS FOUND

Check the BeatStep is plugged in with USB and switched on, then try again. The beat is sent on the drum channel set in SETUP (10 out of the box), which should match the BeatStep's drum sequencer channel.

THE LOOP TAB SAYS INPUT OFF, OR NO INPUTS ARE LISTED

The looper can only use inputs on the output you chose in SETUP. Pick your interface's ASIO driver there (WASAPI outputs have no inputs), then choose the input on the Loop tab.

I HEAR MY MIC TWICE, OR WITH AN ECHO

Your interface is already letting you hear it. Turn down HEAR INPUT (Loop tab, knob 7).

A RECORDING HAS SMALL GAPS

The PC couldn't keep up. Choose a bigger BUFFER in SETUP and close heavy apps while recording.

THE PADS DON'T RUN ACTIONS

Set the BeatStep's Control Mode pads to MIDI channel 16 in MIDI Control Center, or teach PotentialBox your pads in SETUP > PAD ACTIONS.

A SYNTH OR DRUM SOUNDS WRONG

A macro may be turned up: check the Macros tab or tap ZERO ALL MACROS. For a fresh start, RESET ALL TO INIT on the Mix tab.

A MACHINE KIT HAS ODD SOUNDS ON SOME PADS

The sorter goes by file names. Sounds with only numbers in their names fill the remaining pads in order. Use EDIT PAD to swap any pad, then save it in My Kits.

POTENTIALBOX WON'T START

Run the installer again. If it still won't start, the last lines of `%APPDATA%\PotentialBox\log.txt` say why; include them if you report a problem on GitHub.

23 Credits and licence

PotentialBox is made by **PotentialBreaks** and is free and open source under the **MIT licence**.

Built with

PROJECT	USED FOR	LICENCE
Python 3.12	Runs the app	PSF License
NumPy, SciPy	Sound processing	BSD
sounddevice (with PortAudio)	Audio output, including ASIO	MIT
mido, python-rtmidi	MIDI from the BeatStep Pro	MIT
Tcl/Tk	The screen	BSD-style
AKWF-FREE (Adventure Kid Waveforms)	24 of Synth 2's wavetables	CC0 (public domain)

The libraries are downloaded by the installer from their official sources; they are not included in the PotentialBox zip. The AKWF wavetables are included, as permitted by CC0. All drum sounds in the built-in kits are generated by PotentialBox itself.

> Arturia and BeatStep Pro are trademarks of Arturia. ASIO is a trademark of Steinberg Media Technologies GmbH. Machine names in your sample library belong to their owners. PotentialBox is an independent project and is not affiliated with or endorsed by any of them.